

CHAPTER 9

RANDOM SAMPLES

9.1 STATISTICS AND SAMPLING DISTRIBUTIONS

In the previous chapters, we referred to independent random variables $X_1, \dots, X_n$ having the same distribution, as iid (independent, identically distributed) variables. In statistics, we say that such variables constitute a random sample, since they can be thought of as observations (measurements) independently selected from the same population, or resulting from analogous statistical experiments. A random sample provides an important link between the observed data and the distribution in the population from which it has been selected. Most techniques of statistical inference that will be discussed in later chapters will be based on certain functions of random samples. Such functions are called *statistics* if they depend on observations $X_1, \dots, X_n$, but not on parameter(s) of the distribution. We will discuss the properties of the distributions of statistics (called *sampling distributions*).

Before we start, it should be mentioned that there are several important issues related to the properties of a selection process so that it does yield a random sample. There are also other, and sometimes more appropriate, sampling schemes besides the random sample. That will be discussed in more detail in Chapter 10.

Among various statistics meaningful for the purpose of statistical inference is a sample mean

$$\bar{X} = \frac{X_1 + \dots + X_n}{n},$$

an arithmetic average of all measurements, or equivalently their sum

$$T_n = X_1 + \dots + X_n.$$

Among other important statistic is sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad (9.1)$$

sample standard deviation $S = \sqrt{S^2}$, $\min(X_1, \dots, X_n)$ and $\max(X_1, \dots, X_n)$ —the smallest and the largest values in the data, respectively.

The following example illustrates an interesting distributional property:

■ EXAMPLE 9.1

We will determine the distribution of $T_n = X_1 + \dots + X_n$ and of $\bar{X} = (1/n)T_n$ in a random sample $X_1, \dots, X_n$ selected from an $\text{EXP}(\lambda)$ distribution.

SOLUTION. Based on Theorems 7.5.4 and 7.5.5, the moment generating function (mgf) for the sample mean $\bar{X}$ is

$$m_{\bar{X}}(t) = m_{T_n}\left(\frac{t}{n}\right) = \left[m_X\left(\frac{t}{n}\right)\right]^n = \left(1 - \frac{\lambda t}{n}\right)^{-n}, \quad (9.2)$$

From Theorem 7.5.6, it is clear that distributions of T_n and $\bar{X}$ are $\text{GAM}(n, \lambda)$ and $\text{GAM}(n, \lambda/n)$, respectively.

A similar property for the family of normal distributions will be shown in Section 9.2.

Regardless of the distribution in the sample, one can obtain expectations of some sampling distributions as functions of the moments of original distributions as given in the following theorem:

Theorem 9.1.1 *Let $X_1, \dots, X_n$ be a random sample from a distribution with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Then*

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad (9.3)$$

and similarly

$$E(X_1 + \dots + X_n) = n\mu, \quad \text{Var}(X_1 + \dots + X_n) = n\sigma^2. \quad (9.4)$$

Moreover, $E(S^2) = \sigma^2$.

Proof: Formulas (9.3) and (9.4) can be found in Sections 7.4 and 7.7. The proof that $E(S^2) = \sigma^2$ is very simple. Since $E(W^2) = \text{Var}(W) + [E(W)]^2$ for any random variable W , and $\sum (X_i - \bar{X})^2 = \sum X_i^2 - n(\bar{X})^2$, we have

$$\begin{aligned} E\left(\sum X_i^2\right) - nE(\bar{X})^2 &= n[E(X_i^2) - E(\bar{X})^2] \\ &= n[\sigma^2 + \mu^2 - (\sigma^2/n + \mu^2)] = (n-1)\sigma^2. \end{aligned}$$

□

PROBLEMS

9.1.1 Statistic G_k , defined for $k = 1, 2$ as

$$G_k = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n |X_i - X_j|^k,$$

was proposed as a measure of variation by Jordan (1869). Show that $G_2 = 2S^2$, where S^2 is given by formula (9.1).

9.1.2 Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be two random samples from distributions with means μ_1 and μ_2 , respectively, and the same variance σ^2 . **(i)** Find $E(\bar{X} - \bar{Y})$ and $\text{Var}(\bar{X} - \bar{Y})$. **(ii)** Assuming that $\mu_1 = \mu_2$ and both samples are of equal size, find $n = m$ such that $P(|\bar{X} - \bar{Y}| > \sigma/4) \leq 0.05$.

9.1.3 Let $\bar{X}_1, \bar{X}_2, \bar{X}_3$ be sample means in three independent samples of sizes n_1, n_2, n_3 , respectively. Each sample was obtained from the $N(\mu, \sigma^2)$ distribution. Find the distribution of $V_1 = (1/3)(\bar{X}_1 + \bar{X}_2 + \bar{X}_3)$ and $V_2 = w_1\bar{X}_1 + w_2\bar{X}_2 + w_3\bar{X}_3$, where $w_i = n_i/(n_1 + n_2 + n_3)$.

9.1.4 Let X_1, X_2 be a random sample from the $\text{BIN}(1, p)$ distribution. Obtain the probability that the sample mean does not exceed the sample variance.

9.2 DISTRIBUTIONS RELATED TO NORMAL

The distributions that will be introduced in this section are of special importance in statistical inference. Their applications as sampling distributions of statistics obtained in random samples selected from the normal distribution will be discussed in later chapters. We start from the following theorem:

Theorem 9.2.1 Let $X_1, \dots, X_n$ be a random sample from the normal distribution $N(\mu, \sigma^2)$. Then

$$\bar{X} = \frac{X_1 + \dots + X_n}{n} \quad (9.5)$$

and

$$U = \frac{n-1}{\sigma^2} S^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \quad (9.6)$$

are independent random variables with distributions $N(\mu, \sigma^2/n)$ and χ_{n-1}^2 , respectively.

Proof: There exist several ways of proving this theorem. We will use a multidimensional version of the method of finding the density of a function of random variables.

Let us observe that

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2,$$

since $2 \sum_{i=1}^n (x_i - \bar{x})(\bar{x} - \mu) = 2(\bar{x} - \mu) \sum_{i=1}^n (x_i - \bar{x}) = 0$. Consequently, the joint density of $(X_1, \dots, X_n)$ can be written as

$$\begin{aligned} f(x_1, \dots, x_n) &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 - \frac{n}{2\sigma^2} (\bar{x} - \mu)^2 \right\} \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{u}{2\sigma^2} \right\} \exp \left\{ -\frac{n}{2\sigma^2} (\bar{x} - \mu)^2 \right\}. \end{aligned} \quad (9.7)$$

Using transformation

$$x_j = \bar{x} + v_j \sqrt{u}, \quad j = 1, 2, \dots, n, \quad (9.8)$$

we obtain

$$\sum_{j=1}^n v_j = 0, \quad \sum_{j=1}^n v_j^2 = n, \quad (9.9)$$

which means that two of the v_j 's are functions of the remaining ones. We therefore solve (9.9) for v_{n-1} and v_n , obtaining two solutions:

$$v_{n-1} = \frac{A-B}{2}, \quad v_n = \frac{A+B}{2},$$

or

$$v_{n-1} = \frac{A+B}{2}, \quad v_n = \frac{A-B}{2},$$

where

$$A = -\sum_{k=1}^{n-2} v_k, \quad B = \left(2n - 2 \sum_{k=1}^{n-2} v_k^2 - \sum_{k=1}^{n-2} \sum_{l=1}^{n-2} v_k v_l \right)^{1/2}. \quad (9.10)$$

Thus, to each vector $(v_1, \dots, v_{n-2}, \bar{x}, u)$ with $u > 0$, there are two corresponding systems $(x_1, \dots, x_n)$:

$$\begin{aligned} x_j &= \bar{x} + v_j \sqrt{u}, & j &= 1, \dots, n-2 \\ x_{n-1} &= \bar{x} + \frac{A-B}{2} \sqrt{u}, & x_n &= \bar{x} + \frac{A+B}{2} \sqrt{u} \end{aligned} \quad (9.11)$$

and

$$\begin{aligned} x_j &= \bar{x} + v_j \sqrt{u}, & j &= 1, \dots, n-2 \\ x_{n-1} &= \bar{x} + \frac{A+B}{2} \sqrt{u}, & x_n &= \bar{x} + \frac{A-B}{2} \sqrt{u}. \end{aligned} \quad (9.12)$$

The Jacobian of transformation (9.11) is

$$\begin{aligned} J &= \begin{vmatrix} \frac{\partial x_1}{\partial \bar{x}} & \cdots & \frac{\partial x_n}{\partial \bar{x}} \\ \frac{\partial x_1}{\partial u} & \cdots & \frac{\partial x_n}{\partial u} \\ \frac{\partial x_1}{\partial v_1} & \cdots & \frac{\partial x_n}{\partial v_1} \\ \cdots & \cdots & \cdots \\ \frac{\partial x_1}{\partial v_{n-2}} & \cdots & \frac{\partial x_n}{\partial v_{n-2}} \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 & 1 \\ \frac{v_1}{2\sqrt{u}} & \frac{v_2}{2\sqrt{u}} & \cdots & \frac{v_{n-2}}{2\sqrt{u}} & \frac{A-B}{2} \times \frac{1}{2\sqrt{u}} & \frac{A+B}{2} \times \frac{1}{2\sqrt{u}} \\ \sqrt{u} & 0 & \cdots & 0 & \frac{\sqrt{u}}{2} \left(\frac{\partial A}{\partial v_1} - \frac{\partial B}{\partial v_1} \right) & \frac{\sqrt{u}}{2} \left(\frac{\partial A}{\partial v_1} + \frac{\partial B}{\partial v_1} \right) \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \sqrt{u} & \frac{\sqrt{u}}{2} \left(\frac{\partial A}{\partial v_{n-2}} - \frac{\partial B}{\partial v_{n-2}} \right) & \frac{\sqrt{u}}{2} \left(\frac{\partial A}{\partial v_{n-2}} + \frac{\partial B}{\partial v_{n-2}} \right) \end{vmatrix}. \end{aligned}$$

Since for transformation (9.12) the Jacobian is the same, but with the last two columns interchanged, the absolute value of the Jacobian for transformation (9.12) is the same as for transformation (9.11). We have here

$$|J| = u^{(n-3)/2} h(v_1, \dots, v_{n-2}), \quad (9.13)$$

where h is some function whose exact form is not needed for our purposes.

Substitution of new variables into (9.7), and multiplication by (9.13) gives the same joint density of vector $(\bar{X}, U, V_1, \dots, V_{n-2})$ for transformations (9.11) and (9.12), namely

$$g(\bar{x}, u, v_1, \dots, v_{n-2}) = \frac{1}{(\sigma\sqrt{2\pi})^n} u^{(n-3)/2} e^{-u/2\sigma^2} e^{-(n/2\sigma^2)(\bar{x}-\mu)^2} h(v_1, \dots, v_{n-2}). \quad (9.14)$$

Since the density (9.14) can be written as the product of densities

$$c_1 u^{(n-1)/2-1} e^{-u/2\sigma^2} \times c_2 e^{-(n/2\sigma^2)(\bar{x}-\mu)^2} \times c_3 h(v_1, \dots, v_{n-2}),$$

it is now clear that the random variables $U, \bar{X}$, and $(V_1, \dots, V_{n-2})$ are independent. An inspection of terms involving $\bar{x}$ and u shows that after we adjust the constants, $\bar{X}$ has a $N(\mu, \sigma^2/n)$ distribution and U/σ^2 has a chi-square distribution with $n-1$ degrees of freedom. $\square$

The most remarkable fact here is that $\bar{X}$ and U are independent, even though $\bar{X}$ appears explicitly in the definition of U . This independence is characteristic for a normal distribution. We have the following theorem, which we leave without a proof:

Theorem 9.2.2 *If in a random sample $X_1, \dots, X_n$ variables $\bar{X}$ and U are independent, then the distribution of X_i , $i = 1, \dots, n$, is normal.*

We will now introduce the following definition:

Definition 9.2.1 Let Z be a standard normal random variable, and let U be a chi-square distributed random variable independent of Z , with ν degrees of freedom. Then the random variable

$$X = \frac{Z}{\sqrt{U/\nu}} \quad (9.15)$$

is said to have *Student's t distribution with ν degrees of freedom*. $\square$

The density of X will be derived following the steps outlined in Chapter 6. Starting from the joint density of Z and U , we add a “companion” variable Y to X , find the Jacobian of the transformation $(z, u) \rightarrow (x, y)$, and finally, find the density of X as the marginal in the joint density of X and Y .

By the assumption of independence, the joint density of Z and U is

$$f(z, u) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \times \frac{1}{2^{\nu/2}} \Gamma(\nu/2) u^{\nu/2-1} e^{-u/2}, \quad u > 0. \quad (9.16)$$

Selecting $Y = U$ as a companion variable to X , we have $z = x\sqrt{y/m}$ and $u = y$. The Jacobian equals $J = \sqrt{y/m}$; hence the joint density of X and Y is

$$\varphi(x, y) = Q y^{(\nu-1)/2} e^{-(1/2)(1+x^2/\nu)y}, \quad (9.17)$$

where

$$Q = \frac{1}{\sqrt{\pi\nu} 2^{(m+1)/2} \Gamma(\nu/2)}. \quad (9.18)$$

The density (9.17), except for a constant, is a gamma density with parameters $(m+1)/2$ and $(1/2)(1+x^2/m)$. Thus, the marginal density of X is obtained by integration as

$$\begin{aligned} f_\nu(x) &= \int_0^\infty Q y^{(\nu-1)/2} e^{-(1/2)(1+x^2/\nu)y} dy \\ &= \frac{Q \Gamma((\nu+1)/2)}{[(1/2)(1+x^2/\nu)]^{(\nu+1)/2}} = \frac{\Gamma((\nu+1)/2)}{\sqrt{\pi\nu} \Gamma(\nu/2)} \times (1+x^2/\nu)^{-(\nu+1)/2}. \end{aligned} \quad (9.19)$$

It is evident that the density (9.19) of the t_ν distribution is symmetric around $x = 0$. When $\nu = 1$, we have $\varphi(x) = K(1+x^2)^{-1}$, which is the density of the Cauchy distribution (see Example 7.11). Observe that for $\nu = 1$, the Student's t distribution has no mean (hence no higher moments either). For $\nu > 1$ and $k < \nu$, the k th ordinary moment of t_ν can be obtained as (see Problem 9.2.6):

$$E(X^k) = E(Z^k) \left(\frac{\nu}{2}\right)^{k/2} \frac{\Gamma(\frac{\nu-k}{2})}{\Gamma(\frac{\nu}{2})}, \quad (9.20)$$

where Z has $N(0, 1)$ distribution. Consequently, we have $E(X) = 0$ and $\text{Var}(X) = \nu/(\nu-2)$ for variable X with t_ν distribution ($\nu > 2$).

When the number of degrees of freedom, ν , increases, t_ν distribution approaches the standard normal distribution.

Theorem 9.2.3 *Student's t_ν distribution approaches the standard normal distribution $N(0, 1)$ when the number of degrees of freedom ν increases.*

Proof: The density of a t distribution with ν degrees of freedom is proportional to the product

$$\left[\left(1 + \frac{x^2}{\nu} \right)^\nu \right]^{-1/2} \times \left[\left(1 + \frac{x^2}{\nu} \right) \right]^{-1/2}.$$

When $\nu \rightarrow \infty$, the first term converges to $(e^{x^2})^{-1/2} = e^{-x^2/2}$, while the second term converges to 1. To preserve the integrability to 1, the constants must converge to $1/\sqrt{2\pi}$, and the limiting density is standard normal. $\square$

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution. By Theorem 9.2.1, $\bar{X}$ and $\sum_{i=1}^n (X_i - \bar{X})^2$ are independent, with $\sqrt{n}(\bar{X} - \mu)/\sigma$ and $\sum_{i=1}^n (X_i - \bar{X})^2/\sigma^2$ having standard normal and chi-square distribution with $n-1$ degrees of freedom, respectively. Consequently, the ratio

$$\frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2/(n-1)\sigma^2}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2/(n-1)}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad (9.21)$$

has a Student's t distribution with $n-1$ degrees of freedom. As we will see in Chapters 11 and 12, the random variable (9.21) plays an important role in building schemes for inference on μ .

The upper quantile $t_{\alpha, \nu}$ is defined by the relation $P\{X \geq t_{\alpha, \nu}\} = \alpha$, with X having the Student's t distribution with ν degrees of freedom (so that the upper α -quantile is a $(1-\alpha)$ -quantile).

The following definition introduces yet another distribution important in statistical inference:

Definition 9.2.2 If U and V are independent random variables with distributions $\chi_{\nu_1}^2$ and $\chi_{\nu_2}^2$, respectively, then the random variable

$$X = \frac{U/\nu_1}{V/\nu_2}, \quad (9.22)$$

denoted F_{ν_1, ν_2} , has *Snedecor's F distribution with ν_1 and ν_2 degrees of freedom*. $\square$

Let us derive the density of the random variable X . The joint density of U and V , in view of their independence, is

$$\varphi(u, v) = \frac{1}{2^{\nu_1/2} \Gamma(\nu_1/2)} u^{\nu_1/2-1} e^{-u/2} \times \frac{1}{2^{\nu_2/2} \Gamma(\nu_2/2)} v^{\nu_2/2-1} e^{-v/2} \quad (9.23)$$

for $u > 0, v > 0$. With $Y = V$ as a companion variable, the inverse transformation $(x, y) \rightarrow (u, v)$ is $u = (\nu_1/\nu_2)xy$, and $v = y$ so that the Jacobian equals $J = (\nu_1/\nu_2)y$. Thus, after substitution to (9.23), the joint density of X and Y is

$$\psi(x, y) = \frac{\nu_1^{\nu_1/2}}{\nu_2^{\nu_1/2} 2^{(\nu_1+\nu_2)/2} \Gamma(\nu_1/2) \Gamma(\nu_2/2)} x^{\nu_1/2-1} y^{(\nu_1+\nu_2)/2-1} e^{-(1/2)[\nu_1 x/\nu_2 + 1]y}.$$

The marginal density of X equals, again using the fact that we can separate a gamma-type integral for y ,

$$\begin{aligned} \varphi_{\nu_1, \nu_2}(x) &= \int_0^\infty \psi(x, y) dy = \frac{\nu_1^{\nu_1/2} x^{\nu_1/2-1}}{\nu_2^{\nu_1/2} 2^{(\nu_1+\nu_2)/2} \Gamma(\nu_1/2) \Gamma(\nu_2/2)} \\ &\quad \times \int_0^\infty y^{(\nu_1+\nu_2)/2-1} e^{-(1/2)[\nu_1 x/\nu_2 + 1]y} dy \\ &= \frac{\nu_1^{\nu_1/2} x^{\nu_1/2-1} \Gamma((\nu_1 + \nu_2)/2) 2^{(\nu_1+\nu_2)/2}}{\nu_2^{\nu_1/2} 2^{(\nu_1+\nu_2)/2} \Gamma(\nu_1/2) \Gamma(\nu_2/2) [\nu_1 x/\nu_2 + 1]^{(\nu_1+\nu_2)/2}} \\ &= \frac{\Gamma((\nu_1 + \nu_2)/2) \nu_1^{\nu_1/2} \nu_2^{\nu_2/2}}{\Gamma(\nu_1/2) \Gamma(\nu_2/2)} \times \frac{x^{\nu_1/2-1}}{(\nu_1 x/\nu_2 + 1)^{(\nu_1+\nu_2)/2}}. \end{aligned}$$

PROBLEMS

9.2.1 Let $X \sim t_\nu$. Show that X^2 has $F_{1, \nu}$ distribution.

9.2.2 Show that if x_α is the α -quantile of a random variable X with an F_{ν_1, ν_2} distribution, then $1/x_\alpha$ is the $(1 - \alpha)$ -quantile of a random variable with an F_{ν_2, ν_1} distribution.

9.2.3 Let X, Y, W be independent random variables such that $X \sim N(0, 1)$, $Y \sim N(1, 1)$, and $W \sim N(2, 4)$, respectively. Find k such that

$$P \left\{ \frac{X^2 + (Y - 1)^2}{X^2 + (Y - 1)^2 + (W - 2)^2/4} > k \right\} = 0.05.$$

- 9.2.4** Let $X_1 \sim \text{GAM}(1, \lambda)$, $X_2 \sim \text{GAM}(2, \lambda)$, $X_3 \sim \text{GAM}(3, \lambda)$ be independent random variables. Find constant a such that variable $Y = aX_1/(X_2 + X_3)$ has an F distribution.
- 9.2.5** Show that variable $aX/(1 + aX)$, where $X \sim F(\nu_1, \nu_2)$ and $a = \nu_1/\nu_2$, has a $\text{BETA}(\nu_1/2, \nu_2/2)$ distribution.
- 9.2.6** Derive formula (9.20). [*Hint*: Use formula (8.52) for the k th moment of a $\text{GAM}(\alpha, \lambda)$ distribution and the fact that variables X and U in (9.15) are independent.]
- 9.2.7** Derive the formula for $E(X^k)$, where $X \sim F(\nu_1, \nu_2)$. (*Hint*: Use similar approach as in Problem 9.2.6).

9.3 ORDER STATISTICS

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution with cdf F and density f . We will consider random variables $X_{1:n}, X_{2:n}, \dots, X_{n:n}$, where $X_{i:n}$ is the i th in magnitude among $X_1, X_2, \dots, X_n$. Thus

$$X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}. \quad (9.24)$$

Since variables X_i 's are continuous, the probability of two observations being equal is zero, and therefore we can assume that all inequalities in (9.24) are strict.

Definition 9.3.1 The random variables (9.24) are called *order statistics* of the sample $X_1, X_2, \dots, X_n$, and for $k = 1, \dots, n$, $X_{k:n}$ will be referred to as the k th order statistic. $\square$

In particular, the first order statistic, $X_{1:n}$, is the minimum of the sample, while the n th order statistic, $X_{n:n}$, is the maximum. We will derive the joint distribution of all n order statistics, of two selected order statistics, and also the marginal distribution of any order statistic. We will start from the latter. Let G_k be the cdf of order statistic $X_{k:n}$. Then $X_{k:n} \leq t$ if at least k variables in the sample $X_1, X_2, \dots, X_n$ satisfy the inequality $X_i \leq t$. The number of observations in the sample that satisfy $X_i \leq t$ has a binomial distribution with probability of success $p = F(t) = P(X_i \leq t)$, and therefore

$$G_k(t) = P\{X_{k:n} \leq t\} = \sum_{r=k}^n \binom{n}{r} [F(t)]^r [1 - F(t)]^{n-r}. \quad (9.25)$$

Two particular cases deserve attention here. If $k = 1$, we obtain

$$G_1(t) = P\{X_{1:n} \leq t\} = 1 - P\{X_1 > t, \dots, X_n > t\} = 1 - [1 - F(t)]^n,$$

and if $k = n$, then

$$G_n(t) = P\{X_{n:n} \leq t\} = P\{X_1 \leq t, \dots, X_n \leq t\} = [F(t)]^n.$$

Differentiating with respect to t , we obtain densities

$$g_1(t) = n[1 - F(t)]^{n-1}f(t) \quad \text{and} \quad g_n(t) = n[F(t)]^{n-1}f(t). \quad (9.26)$$

To find the density $g_k(t)$, we differentiate (9.25) obtaining

$$\begin{aligned}
 g_k(t) &= \sum_{r=k}^n \binom{n}{r} r [F(t)]^{r-1} f(t) [1 - F(t)]^{n-r} \\
 &\quad - \sum_{r=k}^{n-1} \binom{n}{r} [F(t)]^r (n-r) [1 - F(t)]^{n-r-1} f(t) \\
 &= \sum_{r=k}^n \frac{n!}{(r-1)!(n-r)!} [F(t)]^{r-1} [1 - F(t)]^{n-r} f(t) \\
 &\quad - \sum_{r=k+1}^n \frac{n!}{(r-1)!(n-r)!} [F(t)]^{r-1} [1 - F(t)]^{n-r} f(t).
 \end{aligned}$$

All terms except one cancel in the last two sums.

Theorem 9.3.1 *The density of the k th order statistic is given by*

$$g_k(t) = \frac{n!}{(k-1)!(n-k)!} [F(t)]^{k-1} [1 - F(t)]^{n-k} f(t). \quad (9.27)$$

To find distributions of more complex functions of order statistics, one should start from the joint distribution of all order statistics given in the following theorem:

Theorem 9.3.2 *The joint density of $X_{1:n}, X_{2:n}, \dots, X_{n:n}$ is given by*

$$\phi(y_1, \dots, y_n) = \begin{cases} n! f(y_1) \cdots f(y_n) & \text{if } y_1 < \cdots < y_n \\ 0 & \text{otherwise.} \end{cases} \quad (9.28)$$

Instead of a formal proof let us use some heuristic argument here. A joint density in a random sample is $g(x_1, \dots, x_n) = f(x_1) \cdots f(x_n)$. While observations $X_1, \dots, X_n$ can be arranged in $n!$ ways, only one arrangement has observations ordered in ascending magnitude ($Y_1 < Y_2 < \cdots < Y_n$).

The next theorem gives the density of a joint distribution of any two order statistics.

Theorem 9.3.3 *Let $X_1, \dots, X_n$ be a random sample from a continuous distribution with density f and cdf F . For any $k < l$ and $s < t$, the joint distribution of $X_{k:n}$ and $X_{l:n}$ is given by the formula*

$$\begin{aligned}
 g_{k,l}(s, t) &= \frac{n!}{(k-1)!(l-k-1)!(n-l)!} [F(s)]^{k-1} [F(t) - F(s)]^{l-k-1} \\
 &\quad \times [1 - F(t)]^{n-l} f(s) f(t)
 \end{aligned} \quad (9.29)$$

and $g_{k,l}(s, t) = 0$ otherwise.

Proof: The density (9.29) is obtained by direct integration of the joint density (9.28) with respect to all y_i such that $i < k, k < i < l$, and $i > l$. $\square$

■ EXAMPLE 9.2 Distribution of the Range

We present two different ways of deriving the distribution of a range $R = X_{n:n} - X_{1:n}$ of a random sample $X_1, \dots, X_n$.

SOLUTION 1. The first solution consists of finding the joint distribution of $X_{1:n}$ and $X_{n:n}$, and then determining the distribution of $X_{n:n} - X_{1:n}$ using techniques for transformations of random variables discussed in Chapter 6.

For $s < t$, the direct application of (9.29) gives

$$\begin{aligned} g_{1,n}(s, t) &= \frac{n!}{(0)!(n-2)!(0)!} [F(t) - F(s)]^{n-2} f(s) f(t) \\ &= n(n-1) [F(t) - F(s)]^{n-2} f(s) f(t). \end{aligned}$$

Since $R = X_{n:n} - X_{1:n}$, we can take the transformation $r = t - s$ and its “companion” $w = t$. Solving for t and s , we obtain $s = w - r$ and $t = w$ with $|J| = 1$. Thus, the density of R is the marginal density for $r > 0$,

$$\begin{aligned} h_R(r) &= \int_{-\infty}^{\infty} g_{1,n}(w-r, w) dw \\ &= \int_{-\infty}^{+\infty} n(n-1) [F(w) - F(w-r)]^{n-2} f(w-r) f(w) dw. \end{aligned} \tag{9.30}$$

SOLUTION 2. A more direct solution is to condition on one of the variables, say $X_{1:n}$. If $X_{1:n} = s$, then the probability that the range is less than r is $\{[F(s+r) - F(s)]/[1 - F(s)]\}^{n-1}$, since all remaining observations must fall between s and $s+r$. The density of $X_{1:n}$ is given by (9.26) and therefore

$$\begin{aligned} P\{R \leq r\} &= \int_{-\infty}^{+\infty} \left\{ \frac{[F(s+r) - F(s)]^{n-1}}{[1 - F(s)]^{n-1}} \right\}^{n-1} \times n[1 - F(s)]^{n-1} f(s) ds \\ &= \int_{-\infty}^{+\infty} n[F(s+r) - F(s)]^{n-1} f(s) ds. \end{aligned}$$

Differentiation with respect to r gives the density of R .

It is clear that unlike variables $X_1, \dots, X_n$ in a random sample, order statistics $X_{1:n}, \dots, X_{n:n}$ are dependent variables. For example, $X_{i:n} \leq X_{j:n}$ for any $1 \leq i < j \leq n$.

■ EXAMPLE 9.3

We will obtain correlation ρ of two extreme order statistics, $X_{1:n}$ and $X_{n:n}$, in a random sample from $U[0, \theta]$ distribution.

SOLUTION. To find ρ , we need the first and second moments of both variables $X_{1:n}$ and $X_{n:n}$, and their covariance. Based on Example 5.14 and densities (9.26) we have

$$\begin{aligned} E(X_{1:n}) &= \frac{n}{\theta} \int_0^\theta t \left(1 - \frac{t}{\theta}\right)^{n-1} dt = n\theta \int_0^1 (1-w)w^{n-1} dw \\ &= \frac{1}{n+1} \theta \end{aligned} \tag{9.31}$$

and

$$E(X_{n:n}) = \frac{n}{\theta} \int_0^\theta t \left(\frac{t}{\theta}\right)^{n-1} dt = n\theta \int_0^1 u^n du = \frac{n}{n+1}\theta. \quad (9.32)$$

Similarly, the second moments are

$$\begin{aligned} E(X_{1:n}^2) &= \int_0^\theta t^2 g_1(t) dt = \frac{n}{\theta} \int_0^\theta t^2 \left(1 - \frac{t}{\theta}\right)^{n-1} dt \\ &= n\theta^2 \int_0^1 (1-w)^2 w^{n-1} dw = \frac{2\theta^2}{(n+1)(n+2)} \end{aligned} \quad (9.33)$$

and

$$E(X_{n:n}^2) = \int_0^\theta t^2 g_n(t) dt = \frac{n}{\theta} \int_0^\theta t^2 \left(\frac{t}{\theta}\right)^{n-1} dt = \frac{n\theta^2}{n+2}. \quad (9.34)$$

Consequently,

$$\text{Var}(X_{1:n}) = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)}$$

and

$$\text{Var}(X_{n:n}) = \frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)} = \text{Var}(X_{1:n}).$$

The equality of variances should not be surprising. Since $g_1(t) = g_n(\theta - t)$ for $0 \leq t \leq \theta$, the variability in distributions of $X_{1:n}$ and $X_{n:n}$ is the same.

To find $\text{Cov}(X_{1:n}, X_{n:n})$, we first obtain $E(X_{1:n}X_{n:n})$ using the joint density

$$g_{1,n}(s, t) = \frac{n(n-1)}{\theta^n} (t-s)^{n-2} \quad \text{for } 0 \leq s < t \leq \theta$$

based on formula (9.29). Then

$$E(X_{1:n}X_{n:n}) = \int_0^\theta \int_0^t s t \frac{n(n-1)}{\theta^n} (t-s)^{n-2} ds dt = \frac{\theta^2}{n+2},$$

and consequently

$$\text{Cov}(X_{1:n}, X_{n:n}) = \frac{\theta^2}{n+2} - \frac{\theta}{(n+1)} \times \frac{n\theta}{(n+1)} = \frac{\theta^2}{(n+1)^2(n+2)}. \quad (9.35)$$

Finally,

$$\begin{aligned} \rho &= \frac{\text{Cov}(X_{1:n}, X_{n:n})}{\sqrt{\text{Var}(X_{1:n})} \sqrt{\text{Var}(X_{n:n})}} \\ &= \frac{\theta^2}{(n+1)^2(n+2)} \times \frac{(n+1)^2(n+2)}{n\theta^2} = \frac{1}{n}. \end{aligned} \quad (9.36)$$

The obtained result is intuitive. The correlation of $X_{1:n}$ and $X_{n:n}$ decreases when the number of observations in the sample (sample size) increases.

■ EXAMPLE 9.4 Theory of Outliers

As another illustration of the use of order statistics, we will present basic definitions and results of the theory of outliers proposed by Neyman and Scott (1971).

An outlier is an observation that is sufficiently far away from the remaining observations to justify the suspicion that it results from an observational or recording error, or perhaps from a sudden undetected change of conditions that made this particular observation obey different probability distributions than those obeyed by the remaining observations. What we present here is a theory of right outliers. The results for left outliers may be obtained by obvious modification. Thus, we have a sample, $x_1, \dots, x_n$, with values of order statistics $x_{1:n}, \dots, x_{n:n}$ where $x_{n:n}$ is, in some sense, “too far away” from the rest of observations. The intuition of “being too far away” has been formalized in a number of ways in various approaches to the detection of outliers, typically in terms of relation of $x_{n:n}$ to the average and standard deviation of the sample.

Neyman and Scott (1971) proposed the definition based on order statistics. In what follows, we will always assume that $n \geq 3$ and $r > 0$.

Definition 9.3.2 The sample $x_1, \dots, x_n$ contains an (r, n) right outlier, if

$$x_{n:n} - x_{n-1:n} > r(x_{n-1:n} - x_{1:n}). \quad (9.37)$$

We will simply use the term *outlier* for the right outlier. □

It is clear that if only $x_{n:n} \neq x_{n-1:n}$, the sample contains an (r, n) outlier for all r satisfying the condition

$$r < \frac{x_{n:n} - x_{n-1:n}}{x_{n-1:n} - x_{1:n}}.$$

Intuitively, the *outlier* in the usual sense is an (r, n) outlier for r large enough.

Theorem 9.3.4 A random sample $X_1, \dots, X_n$ from a continuous distribution with cdf F and density f will contain an (r, n) outlier with probability

$$\pi(r, n; F) = n(n-1) \int_{-\infty}^{\infty} \int_x^{\infty} \left[F\left(\frac{y+rx}{r+1}\right) - F(x) \right]^{n-2} f(y) f(x) dy dx. \quad (9.38)$$

Proof: Indeed, if we condition on $x_{1:n} = x$ and $x_{n:n} = y > x$, then the sample will contain an (r, n) outlier if (9.37) holds, hence if

$$x < X_{n-1:n} < \frac{y+rx}{r+1}.$$

This event, in turn, occurs if $n-2$ remaining observations lie between x and $(y+rx)/(r+1)$, which occurs with probability $[F((y+rx)/(r+1)) - F(x)]^{n-2}$. Multiplying by the joint density of $(X_{1:n}, X_{n:n})$ and integrating, we obtain (9.38). □

Formula (9.38) allows one to compute the probability that an (r, n) outlier will appear for a given F and n . One of the essential features that characterize situations in statistics is that the distribution F that governs the selection of the sample is not known. Typically, we know only that the true F belongs to a certain family $\mathcal{F}$ of distributions. For example, the experimenter may know that the sample she observes was obtained from a normal distribution but not know the value(s) of the parameter(s).

Accordingly, we introduce the following definitions:

Definition 9.3.3 The family $\mathcal{F}$ of distributions is (r, n) *outlier resistant* if

$$\sup_{F \in \mathcal{F}} \pi(r, n; F) < 1 \quad (9.39)$$

and is (r, n) *outlier prone* if

$$\sup_{F \in \mathcal{F}} \pi(r, n; F) = 1. \quad (9.40)$$

Moreover, a family $\mathcal{F}$ of distributions is *totally outlier resistant* if condition (9.39) holds for all $r > 0$ and $n \geq 3$; it is called *totally outlier prone*, if condition (9.40) holds for all $r > 0$ and $n \geq 3$. $\square$

Among other results proved by Neyman and Scott (1971), we mention two theorems. One of them asserts that *the family of all normal distributions is totally outlier resistant*. The other asserts that *the family of all gamma distributions is totally outlier prone*.¹⁶ The practical consequences of these two theorems are as follows: On the one hand, if we have a sample $x_1, \dots, x_n$ from a normal distribution, we can find the largest r for which this sample contains an (r, n) outlier. This r is equal to the ratio $r^* = (x_{n:n} - x_{n-1:n}) / (x_{n-1:n} - x_{1:n})$. We can then find the quantity (9.38) for $\mathcal{F}$ being a family of the normal distribution and $r = r^*$. If this quantity is sufficiently small, we have good reason to suspect that the element $x_{n:n}$ in the sample is a genuine outlier, in the sense of representing observations from a distribution other than that of the rest of the sample. This gives a practical procedure for the rejection of outliers.

On the other hand, if we have a sample $x_1, \dots, x_n$ from a gamma distribution, we can *never* reject the largest element as an outlier if our decision is to be based on the observed values only. To see this, suppose that we have data such as $x_{1:4} = 0.5$, $x_{2:4} = 0.55$, $x_{3:4} = 1$, and $x_{4:4} = 1,000,001$. We cannot reject the observation 1,000,001 as an outlier, since the probability of a configuration with the ratio

$$\frac{x_{4:4} - x_{3:4}}{x_{3:4} - x_{1:4}} = 2,000,000$$

or more has a probability arbitrarily close to 1 for *some* gamma distribution. More precisely, for every $\epsilon > 0$, there is a gamma distribution such that the probability of the configuration above exceeds $1 - \epsilon$.

Our conclusion that in practice one should never reject any sample element as an outlier if the sample is known to come from some gamma distribution is actually counterintuitive. Such a conclusion does unexpectedly provide a powerful argument for the Bayesian approach to statistics (which will be discussed in Chapter 16), but the point is that it seldom, if ever, happens that the statistician knows *only* that the sample comes from a gamma distribution. He usually has some idea about the parameters, based on his experience, imagination, understanding of the situation, and so on. Such knowledge, vague as it may be, allows the statistician to regard some gamma distributions in the family $\mathcal{F}$ as “more plausible” than others, and perhaps even eliminate some members of $\mathcal{F}$ as impossible in the given situation. Now, if we restrict the family $\mathcal{F}$ to *some* gamma distributions only (by putting a bound on parameter α), then $\mathcal{F}$ is no longer totally outlier prone, and rejection of outliers becomes justifiable.

¹⁶For further development of the suggested theory of outliers, see Green (1974, 1976).

PROBLEMS

- 9.3.1** In a random sample $X_1, \dots, X_n$ selected from $\text{EXP}(\lambda)$ distribution find the density of: **(i)** $X_{1:n}$. **(ii)** $\sqrt{n}X_{1:n}$. **(iii)** $X_{2:n}$.
- 9.3.2** Let $X_1, \dots, X_n$ be a random sample selected from the $U[0, 1]$ distribution. **(i)** Find the distribution of variable $W = X_{i:n} - X_{i-1:n}$, $1 < i \leq n$. **(ii)** Show that the distribution of $X_{j:n}$ is $\text{BETA}(j, n-j+1)$ for $j = 1, \dots, n$. **(iii)** Find the distribution of a sample median $X_{k+1:n}$, and obtain its variance if $n = 3, 7, 15, 2k+1$. Do you see any trend?
- 9.3.3** Let X_1, X_2 be a random sample of size 2 from a continuous distribution with a median θ . **(i)** Find $P(X_{1:2} < \theta < X_{2:2})$. **(ii)** Generalize part **(i)** finding $P(X_{1:n} < \theta < X_{n:n})$ in a random sample of size n .
- 9.3.4** Use results from Example 9.3 to determine $\text{Var}(X_{l:n} - X_{k:n})$, $l > k$, in a random sample $X_1, \dots, X_n$ from the $U[0, \theta]$ distribution.
- 9.3.5** Let $X_1, \dots, X_n$ be a random sample from the $U[0, 1]$ distribution. Find sample size n such that $E(R) = 0.75$, where $R = X_{n:n} - X_{1:n}$.
- 9.3.6** Let $X_1, \dots, X_n$ be a random sample from the $U(0, \theta)$ distribution. Find: **(i)** $E(X_{1:n}^m)$ and $E(X_{n:n}^m)$. **(ii)** $E(X_{n:n}|X_{1:n})$. **(iii)** The cdf of variable $V = X_{n:n} - X_{1:n}$. **(iv)** The probability that the interval $[X_{1:n}, X_{n:n}]$ does not include a given point x_0 . **(v)** The point x_0 for which the probability in part (iv) is the smallest.
- 9.3.7** Let $X_1, \dots, X_5$ be a random sample from the $\text{BETA}(2, 1)$ distribution. **(i)** Find the density of a joint distribution of $X_{1:5}, X_{2:5}, X_{4:5}$. **(ii)** Obtain $E(X_{2:5}|X_{4:5})$. **(iii)** Obtain $E(X_{4:5}|X_{2:5})$. **(iv)** Find the distribution of $Y = X_{2:5}/X_{1:5}$. **(v)** Check if variables $X_{1:n}/X_{n:n}$ and $X_{n:n}$ are independent.
- 9.3.8** Let X_1, X_2 be a random sample from the $U(0, 1)$ distribution. Find the probability that the values in the sample divide the $(0, 1)$ interval into three parts, each of the length at least 0.3 (i.e., $X_{1:2} > 0.3$, $X_{2:2} > X_{1:2} + 0.3$, and $X_{2:2} < 0.7$).
- 9.3.9** Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ be two independent random samples from the same continuous distribution with a density f . Show that $P\{X_{i:n} \leq t\} \geq P\{Y_{j:n} \leq t\}$ for every t if and only if $i \leq j$.
- 9.3.10** Find $\pi(r, n; F_\theta)$ when F_θ is the $U[0, \theta]$ distribution. Find $\pi(r, n; \mathcal{F})$ for the family $\mathcal{F} = \{\mathcal{F}_\theta\}$, $\theta > 0$.

9.4 GENERATING RANDOM SAMPLES

In various quantitative problems that are too complex to be studied mathematically, one can often use simulations to find an approximate solution. Such techniques of statistical sampling, known today as Monte Carlo methods,¹⁷ have been used in a very simple form (e.g., tossing dice) for centuries, but became a widely applicable formal methodology only with the invention of electronic computers in the middle of the twentieth century.

¹⁷Stanisław Ulam, a Polish born American mathematician working with John von Neuman and Nicholas Metropolis on the Manhattan Project during World War II, developed computer algorithms for statistical sampling. He also worked on transforming deterministic problems into random ones that could be solved by simulations. In his spare time Ulam investigated probabilities of winning a card game of solitaire, and that is how the new methodology got named after a famous casino.

Since then, with increasing computing power and easy access to technology, simulations have become an important and powerful tool of modern statistics. Today complex statistical problems are often investigated based on generated random samples from specific distributions that are easy to obtain even using available statistical software packages. We now introduce related concepts and explain the theory behind the Monte Carlo simulation process.

To generate a random sample from any distribution, we first need to generate independent observations from the $U[0, 1]$ distribution. Over the last 40 years, several algorithms have been proposed and extensively tested, so the ones that we use today are of really good quality. However, “random number generators” are not really generators of random numbers. They produce numbers that can be periodical but with periods much larger than we are able to detect, and therefore not affecting the quality of our simulation process. The interested readers should consult some additional literature, such as Ross (2006).

We will now discuss the random sample generation process, assuming that a good random generator of a $U[0, 1]$ distribution is available.

The following example will show how one can generate a random sample from a simple discrete distribution:

■ EXAMPLE 9.5

Let X be a Bernoulli random variable such that

$$P(X = 1) = 1 - P(X = 0) = p = 0.2.$$

To generate a single observation from this distribution, we must first generate U , a single observation from the $U[0, 1]$ distribution, and then transform it into X observation in a following way:

$$X = \begin{cases} 1 & \text{if } U \leq 0.2 \\ 0 & \text{otherwise.} \end{cases} \quad (9.41)$$

To generate a random sample of size n , we simply need to generate a random sample $U_1, \dots, U_n$ from the $U[0, 1]$ distribution and then transform each U_i into X_i in the way explained above. The transformation given by (9.41) is not the only one possible here. Other possibilities could, for example, be

$$X = \begin{cases} 1 & \text{if } U > 0.8 \\ 0 & \text{otherwise} \end{cases} \quad \text{or} \quad X = \begin{cases} 1 & \text{if } 0.3 < U \leq 0.5 \\ 0 & \text{otherwise,} \end{cases}$$

as long as $P(X = 1) = 0.2$.

To generate a random sample $Y_1, \dots, Y_n$ from the binomial distribution $\text{BIN}(k, p)$, we will need $k \times n$ observations from the $U[0, 1]$ distribution. Then, we will transform each of them into the $\text{BIN}(1, p)$ according to (9.41) and obtain $Y_1, \dots, Y_n$ as

$$Y_i = \sum_{j=(i-1) \times n + 1}^{i \times n} X_j$$

for $i = 1, \dots, k$. Random samples from any distribution related to Bernoulli trials can be generated in a similar way.

The next example shows how to generate values from other discrete distributions.

■ EXAMPLE 9.6

Let X be a random variable with a $\text{POI}(\lambda)$ distribution, in which $P(X = x) = e^{-\lambda} \lambda^x / x!$. After a value of random variable U is generated from $U[0, 1]$, the value x of

X is determined in a following way: x equals 0 when $U \leq e^{-\lambda}$, otherwise x is the only positive integer for which the inequality

$$\sum_{i=0}^{x-1} \frac{e^{-\lambda} \lambda^i}{i!} < U \leq \sum_{i=0}^x \frac{e^{-\lambda} \lambda^i}{i!}$$

holds.

To generate observations from continuous distributions with a cdf F such that F^{-1} has a closed form, we can apply Theorem 5.4.2, as illustrated by the following example:

■ EXAMPLE 9.7

To obtain a random sample $X_1, \dots, X_n$ from the $\text{EXP}(\lambda)$ distribution, we have to generate a random sample $U_1, \dots, U_n$ from $U[0, 1]$ and for any $i = 1, \dots, n$, take

$$X_i = -\frac{1}{\lambda} \log(1 - U),$$

since $F_X^{-1}(u) = -\frac{1}{\lambda} \log(1 - u)$ is an inverse of the cdf $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.

The generation that we just presented can also serve as a first step in generating some distributions from a gamma family. Variable $Y = X_1 + \dots + X_k$, where variables X_i 's are independent and have the same $\text{EXP}(\lambda)$ distribution, will, by Theorem 8.4.1, have a $\text{GAM}(k, \lambda)$ distribution. Therefore, to generate $Y_1, \dots, Y_n$, we could start by generating a random sample $U_i, \dots, U_{n \times k}$ from $U[0, 1]$ and then transform it into a sample $X_1, \dots, X_{n \times k}$ to finally obtain a random sample $Y_1, \dots, Y_n$, where for any $i = 1, \dots, n$,

$$Y_i = \sum_{j=(i-1) \times k + 1}^{i \times k} X_j.$$

Unfortunately, this approach can be used only if the shape parameter in a gamma distribution is a positive integer.

In general, there exist methods that can be used to generate all probability distributions with an invertible cdf that is not in a closed form. One of these techniques, called the Accept-Reject algorithm, is presented below.

The idea here is that the two observations are generated independently. One observation, U , is from the $U[0, 1]$ distribution, and the other, Y , is from a selected continuous distribution, which is easy to generate. If a certain condition is satisfied, the Y observation is “accepted” as an X observation and included in a random sample $X_1, \dots, X_n$. Otherwise, it is “rejected,” and the next pair (U, Y) is generated. Pairs (U, Y) continue to be generated until there are n observations $X_1, \dots, X_n$ in the sample.

To introduce the condition that makes some of the Y observations become X observations, and to explain why the process is really valid, we start with finding a constant C , such that

$$f(x) \leq Cg(y), \quad (9.42)$$

where f is the density of a distribution we are interested in (a “target” distribution), and g is a density of an easy to generate “companion” distribution. The generated Y observation is “accepted” as an X observation if

$$U \leq \frac{f(Y)}{Cg(Y)}; \quad (9.43)$$

otherwise, it is “rejected.” To validate the algorithm, we have to show that the distribution of X is the same as a conditional distribution of Y under the condition (9.43) so that

$$F_X(y) = F_Y \left(y \mid U \leq \frac{f(Y)}{Cg(Y)} \right).$$

Since

$$\begin{aligned} P \left(U \leq \frac{f(Y)}{Cg(Y)} \right) &= \int_{-\infty}^{+\infty} P \left(U \leq \frac{f(y)}{Cg(y)} \right) g(y) dy \\ &= \int_{-\infty}^{+\infty} \frac{f(y)}{Cg(y)} g(y) dy = \frac{1}{C} \int_{-\infty}^{\infty} f(y) dy = \frac{1}{C}, \end{aligned} \quad (9.44)$$

we have

$$\begin{aligned} P \left(Y \leq y \mid U \leq \frac{f(Y)}{Cg(Y)} \right) &= \frac{P(Y \leq y, U \leq f(Y)/Cg(Y))}{P(U \leq f(Y)/Cg(Y))} \\ &= C \int_{-\infty}^y g(t) \int_0^{f(t)/Cg(t)} du dt \\ &= C \int_{-\infty}^y g(t) \frac{f(t)}{Cg(t)} dt = \int_{-\infty}^y f(t) dt = F_X(y) \end{aligned}$$

as was to be shown.

Two questions still need to be answered here: Does it matter how the constant C is selected? And how many (U, Y) pairs will have to be generated before we get $X_1, \dots, X_n$? From (9.44), it is clear that although any $C < \infty$ that satisfies (9.42) can be used, the optimal choice is $C = \sup_y f_Y(y)/g(y)$. Unfortunately, for some distributions such $C < \infty$ cannot be found. One could then use more advanced methods (see e.g., Ross, 2006).

PROBLEMS

- 9.4.1** Obtain a sample of size 6 from a POI(2) distribution based on following six independent observations from $U[0, 1]$ distribution: 0.090907, 0.185040, 0.124341, 0.299086, 0.428996, 0.927245.
- 9.4.2** Obtain a random sample of size 4 from a Pareto distribution with a density $f(x) = (1+x)^{-2}$ for $x > 0$ and 0 otherwise. Use the following random sample from $U[0, 1]$: 0.187724, 0.386997, 0.182338, and 0.028113.
- 9.4.3** The double exponential (or Laplace) distribution has a density given by the formula $f(x) = (\lambda/2)e^{-\lambda|x|}$ for $-\infty < x < \infty, \lambda > 0$. Obtain a random sample from the Laplace distribution with $\lambda = 2$ based on a random sample 0.744921 and 0.464001 from the $U[0, 1]$ distribution.
- 9.4.4** A generalized Laplace distribution has a density given by the formula

$$f(x) = \begin{cases} p\lambda_1 e^{-\lambda_1 x} & \text{if } x \geq 0 \\ (1-p)\lambda_2 e^{\lambda_2 x} & \text{otherwise,} \end{cases} \quad (9.45)$$

where $\lambda_1 > 0, \lambda_2 > 0$. Generate two independent observations from a generalized Laplace distribution with $p = 1/4, \lambda_1 = 3$, and $\lambda_2 = 1/2$, based on a random sample 0.647921, 0.049055 from $U[0, 1]$ distribution.

9.4.5 Generate a random sample from the Gompertz distribution with survival function $S(t) = \exp\{1 - \exp(2t)\}$ using the following random sample from the $U[0,1]$ distribution: 0.289365, 0.228349, 0.732889.

9.4.6 Apply an Accept/Reject method to the Laplace distribution with density $f(x) = 1.5e^{-3|x|}$ to generate observations from a standard normal distribution. List the obtained values and specify how many of them you were able to obtain using the random sample of size 5 from $U[0, 1]$: 0.222795, 0.516174, 0.847152, 0.466449, 0.914370. Use the optimal choice of C .

9.4.7 The Box–Muller transformation of two independent, uniform variables into two independent standard normal variables was presented in Theorem 6.4.1. Another algorithm, proposed by Marsaglia and Bray (1964), is to generate U_1 and U_2 as two independent observations from $U[-1, 1]$. If $V = U_1^2 + U_2^2 \leq 1$, then define

$$Z_1 = U_1 \sqrt{\frac{-2 \log V}{V}} \quad \text{and} \quad Z_2 = U_2 \sqrt{\frac{-2 \log V}{V}};$$

otherwise, start the procedure again. Show that Z_1 and Z_2 are independent and have an $N(0, 1)$ distribution.

9.4.8 Kennedy and Gentle (1980) provide the following algorithm for generating a beta distribution: Generate U_1 and U_2 —two independent observations from the $U[0, 1]$ distribution. For $\alpha > 0$ and $\beta > 0$ denote $V_1 = U_1^\alpha$ and $V_2 = U_2^\beta$. According to the Accept/Reject algorithm, let $X = V_1/(V_1 + V_2)$ if $V_1 + V_2 \leq 1$; otherwise, start the procedure again. **(i)** Determine the distribution of a random variable X . **(ii)** Use this algorithm to generate a random sample of size 3 from $BETA(0.732, 1.281)$.

9.4.9 Genest (1987) provides the following algorithm for generating random samples from the so-called Frank family of bivariate distributions: (a) Generate two independent observations U_1 and U_2 from $U[0, 1]$. (b) Obtain $T = \alpha^{U_1} + (\alpha - \alpha^{U_1})U_2$. (c) Let $X = U_1$ and $Y = \log_\alpha[T/(T + (1 - \alpha)U_2)]$, where $\alpha > 0, \alpha \neq 1$. **(i)** Show that the bivariate cdf of Frank's distribution has the following form:

$$H_\alpha(x, y) = P(X \leq x, Y \leq y) = \log_\alpha \left\{ 1 + \frac{(\alpha^x - 1)(\alpha^y - 1)}{\alpha - 1} \right\}.$$

(ii) Generate one observation of (X, Y) based on two independent observations from $U[0, 1]$: 0.548291 and 0.179112. Use $\alpha = 4$.

9.5 CONVERGENCE

Several limit theorems were already encountered in past chapters. On several occasions, we looked at the differences between sampling with and without replacement, and noted that these two schemes of sampling become closer one to another “as the population becomes larger.” We also proved the Poisson approximation theorem, which asserts that as n becomes larger and p becomes smaller, the binomial and Poisson probabilities of the same events become close.

The common feature of these theorems was that in each case the probabilities of certain events—or, more generally, distributions of certain random variables—approached some limits with the appropriate change of one or more parameters. Typically, the parameter is an index of some sequence, such as sample size n , but the case of Poisson approximation shows that it can also be a simultaneous change of two parameters that drives the probabilities to their limiting values.

In addition to the limit theorem mentioned above, we have encountered a different kind of limit theorem, exemplified by the law of large numbers (Example 7.55). There we had convergence not only of the distributions but of the random variables themselves.

To grasp the difference between those two classes of situations, observe that one can study convergence of a sequence of distributions $\{F_n\}$ without considering any random variables. On the other hand, if $X_1, X_2, \dots$ is a sequence of independent random variables with the same distribution F , then the sequence $\{F_n\}$ of their distributions clearly converges to F , but we cannot expect any regularity in behavior of the sequence $\{X_n\}$.

In what follows, we consider a sequence $\xi_1, \xi_2, \dots$ of random variables defined on the same sample space $\mathcal{S}$ with σ -field $\mathcal{F}$ of events, and probability measure P on $\mathcal{S}$. Our first objective will be to distinguish various possible modes in which sequence $\{\xi_n\}$ can converge, and discuss their interrelationships. To connect the analysis with subsequent statistical concepts, it is worthwhile to start from some examples of sequences $\{\xi_n\}$ that arise in statistical research and practice.

■ EXAMPLE 9.8

One of the most common situations in statistics is when we have simple random samples. This means that we observe the beginning of a sequence of iid random variables $X_1, X_2, \dots$. Depending on the goal of analysis, given the observations $X_1, \dots, X_n$, the statistician computes the value of some statistic $\xi_n = H(X_1, \dots, X_n)$ and uses ξ_n as means of inference. The behavior of the sequence $\{\xi_n\}$ as n increases tells the statistician to what extent it would be worthwhile to increase the sample size n .

We begin with the definition that captures the type of convergence encountered in the law of large numbers (Theorem 7.8.3).

Definition 9.5.1 The sequence $\{\xi_n\}$ *converges in probability to a constant* c if for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P\{|\xi_n - c| \geq \epsilon\} = 0. \quad (9.46)$$

More generally, we say that $\{\xi_n\}$ *converges in probability to a random variable* ξ if for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P\{|\xi_n - \xi| \geq \epsilon\} = 0. \quad (9.47)$$

Convergence in probability, especially to a constant, is often called a *stochastic convergence*. $\square$

The meaning of (9.46) is that as n increases, it becomes less and less likely that ξ_n will deviate from c by more than ϵ . In (9.47), the interpretation is the same, except that the constant c is replaced by a random quantity. So written more explicitly, (9.47) reads

$$\lim_{n \rightarrow \infty} P\{s : |\xi_n(s) - \xi(s)| \geq \epsilon\} = 0.$$

We will use the symbol $\xrightarrow{P}$ to denote convergence in probability.

■ EXAMPLE 9.9 Laws of Large Numbers

The law of large numbers proved in Section 7.8 asserts convergence in probability of empirical frequencies of an event. Such theorems (asserting convergence in probability of averages of random variables) are called *weak* laws of large numbers (WLLN), to distinguish them from *strong* laws of large numbers (SLLN).

For instance, in the case of a Bernoulli distribution we have the following rephrasing of Theorem 7.8.3:

Theorem 9.5.1 *If S_n has binomial distribution $\text{BIN}(n, p)$, then*

$$\frac{S_n}{n} \xrightarrow{P} p.$$

A stronger type of convergence of sequences of random variables is given in the following definition:

Definition 9.5.2 Let $\xi_1, \xi_2, \dots$ be a sequence of random variables defined on some probability space $(S, \mathcal{F}, \mathcal{P})$. If $\lim \xi_n(s) = \xi(s)$ exists for all points $s \in U$ where $P(U) = 1$, then we say that ξ_n converges to ξ almost everywhere (a.e.), almost surely (a.s.), or with probability 1. $\square$

The following theorem is given without proof:

Theorem 9.5.2 *If $\xi_n \rightarrow \xi$ a.s., then $\xi_n \xrightarrow{P} \xi$.*

The converse to Theorem 9.5.2 is not true, as illustrated by the next example.

■ EXAMPLE 9.10

Let $\xi_1, \xi_2, \dots$ be independent random variables, such that $P\{\xi_n = 1\} = 1/n$ and $P\{\xi_n = 0\} = 1 - 1/n$. Thus, if $0 < \epsilon < 1$, then $P\{|\xi_n| \geq \epsilon\} = P\{\xi_n = 1\} = 1/n \rightarrow 0$, which shows that the sequence $\{\xi_n\}$ converges to 0 in probability. For any sample point, the sequence $\{\xi_n(s)\}$ is simply a sequence of 0's and 1's, and in order to converge to 0, there must be only a finite number of terms equal 1 (i.e., all terms must be 0, starting from some N). But letting $A_n = \{\xi_n = 1\}$, we have $\sum P(A_n) = \sum 1/n = \infty$, and by the second Borel–Cantelli lemma (Theorem 4.5.5), we know that with probability 1 infinitely many events A_n will occur. Thus, $P\{\xi_n \text{ converges}\} = 0$, which shows that convergence in probability does not imply a.s. convergence.

Theorem 9.5.3 *The sequence $\{\xi_n\}$ of random variables ξ converges a.s. to a random variable ξ if and only if for every $k = 1, 2, \dots$,*

$$\lim_{N \rightarrow \infty} P \left\{ |\xi_n - \xi| \geq \frac{1}{k} \text{ for some } n \geq N \right\} = 0, \quad (9.48)$$

or equivalently,

$$\lim_{N \rightarrow \infty} P \left\{ \sup_{n \geq N} |\xi_n - \xi| \geq \frac{1}{k} \right\} = 0.$$

We will now use (9.48), together with the Kolmogorov inequality (Theorem 7.8.5), to prove the sufficiency part of Kolmogorov's three series theorem. This theorem provides conditions for the a.s. convergence of the series $\sum_{j=1}^{\infty} X_j$ of independent random variables $X_1, X_2, \dots$

We will also introduce the method of truncation, a powerful technique of handling limits of sequences of random variables. If X is a random variable and $c > 0$, we define the

truncation of X at c as a random variable Y , defined by

$$Y = \begin{cases} X & \text{if } |X| \leq c \\ 0 & \text{if } |X| > c. \end{cases}$$

Observe that Y is a bounded random variable, so $E(Y)$ and $\text{Var}(Y)$ both exist. We have

Theorem 9.5.4 (Kolmogorov Three Series Theorem) *Let $X_1, X_2, \dots$ be a sequence of independent random variables, and let Y_n be the truncation of X_n at level $c > 0$. Then, $\sum_{n=1}^{\infty} X_n$ converges a.s. if and only if for some $c > 0$,*

$$(a) \sum_{n=1}^{\infty} P\{|X_n| > c\} < \infty.$$

$$(b) \sum_{n=1}^{\infty} E(Y_n) < \infty.$$

$$(c) \sum_{n=1}^{\infty} \text{Var}(Y_n) < \infty.$$

Proof: As mentioned, only the sufficiency of conditions (a)–(c) will be shown; the proof of necessity is beyond the scope of this book.

Let us fix N, k , and $n > N$, and consider the sums

$$\sum_{j=N}^r (Y_j - E(Y_j))$$

for $r = N, \dots, n$. By the Kolmogorov inequality (7.57), we have

$$P \left\{ \max_{N \leq r \leq n} \left| \sum_{j=N}^r (Y_j - E(Y_j)) \right| \geq \frac{1}{k} \right\} \leq \frac{\sum_{j=N}^n \text{Var}(Y_j)}{(1/k)^2}.$$

Letting $n \rightarrow \infty$, we have

$$P \left\{ \sup_{N \leq r} \left| \sum_{j=N}^r (Y_j - E(Y_j)) \right| \geq \frac{1}{k} \right\} \leq k^2 \sum_{n=N}^{\infty} \text{Var}(Y_n).$$

In view of (c), the right-hand side converges to 0 as $N \rightarrow \infty$ for every fixed k . By Theorem 9.5.3, the sequence $\sum_{i=1}^n (Y_i - E(Y_i))$, $n = 1, 2, \dots$, converges a.s. Since $\sum_{i=1}^r E(Y_i)$ converges (condition b), we infer that $\sum_{i=1}^{\infty} Y_i$ converges a.s. To complete the proof, observe that $P\{|X_n| > c\} = P\{X_n \neq Y_n\}$. In view of condition (a) and the first Borel–Cantelli lemma (Theorem 2.6.2), with probability 1 only finitely many terms Y_n will differ from terms X_n . Consequently, $\sum X_n$ and $\sum Y_n$ will a.s. differ only by a finite quantity, and since $\sum Y_n$ converges, so does $\sum X_n$. $\square$

■ EXAMPLE 9.11

For the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \dots = \infty$. With alternating signs, we have an alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \log 2$.¹⁸ What if the signs are allocated at random, by a flip of a fair coin? In other words, we ask about convergence of the series

$$X_1 + X_2 + X_3 + \dots,$$

¹⁸We remind the readers that throughout this text $\log x = \log_e x$.

where X_n assumes values $\pm 1/n$ with probability $1/2$ and X_n 's are independent. Taking $c = 2$, say, we have $Y_n = X_n$ for all n , and all terms of the series (a) in Theorem 9.5.4 are zero. Next we have $E(Y_n) = 0$ and $\text{Var}(Y_n) = E(Y_n^2) = 1/n^2$. Thus, all three series (a)–(c) converge; hence $\sum X_n$ converges a.s. In other words, the probability is zero that random signs $+$ and $-$ in the harmonic series will come out so unbalanced as to make the series diverge.

Observe that this is not only the question of “balancing” the numbers of positive and negative signs. Indeed, the series $\sum 1/\sqrt{n}$ diverges, and the series $\sum \epsilon_n/\sqrt{n}$, where ϵ_n 's are independent and $\epsilon_n = \pm 1$ with probability $1/2$, also diverges a.s.—since in this case the series (c) of variances diverges.

The third important type of convergence is the following definition:

Definition 9.5.3 Let $\xi_0, \xi_1, \xi_2, \dots$ be a sequence of random variables, and let $F_n(t) = P\{\xi_n \leq t\}$, $n = 0, 1, 2, \dots$, be their cdf's. The sequence $\{\xi_n\}$ converges in distribution to ξ_0 if

$$\lim_{n \rightarrow \infty} F_n(t) = F_0(t)$$

for every t at which $F_0(t)$ is continuous. In this case, we can write $\xi_n \xrightarrow{d} \xi_0$. Alternatively, we will use the symbol $F_n \Rightarrow F_0$. $\square$

Before presenting the theorems characterizing convergence in distribution, we will make some comments, that can help clarify the motivation and intention of this concept.

First, observe that convergence in a distribution does not imply anything about the behavior of random variables. For instance, if the variables $\xi_0, \xi_1, \dots$ are iid, then $F_n(t) \equiv F_0(t)$, so $\xi_n \xrightarrow{d} \xi_0$, but we cannot expect any regularity in behavior of the observed values of random variables. Since we require only convergence of the distribution functions, we do not need to have any specific random variables ξ_n in mind. This is the reason behind the dual notation in Definition 9.5.3.

The second question concerns the reasons for requiring the convergence of $F_n(t)$ to $F_0(t)$ only at points of continuity of F_0 (and not at all points). The explanation has to do with the special role played by discontinuities of cdf's. Consider a sequence of degenerate random variables (i.e., constants), defined as $\xi_n = 1/n$, $n = 1, 2, \dots$, and $\xi_0 = 0$. Obviously here $\lim \xi_n = \xi_0$ in the “usual” sense of convergence of observed values of ξ_n to ξ_0 .

We have here

$$F_n(t) = P\{\xi_n \leq t\} = \begin{cases} 0 & \text{if } t < 1/n \\ 1 & \text{if } t \geq 1/n \end{cases}$$

and

$$F_0(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0. \end{cases}$$

For any $t \neq 0$ (where F_0 is continuous), we have $\lim F_n(t) = F_0(t)$. However, at $t = 0$ we have $F_0(0) = 1$, while $F_n(0) = 0$ for all n . Observe that if we put $\xi_n = -1/n$, then $F_n(t)$ will converge to $F_0(t)$ at all points t , including $t = 0$.

The following theorems connect the three types of convergence introduced above, and also some criteria for convergence in a distribution.

Theorem 9.5.5 (Slutsky) If ξ_n and η_n are sequences of random variables such that $\xi_n - \eta_n \xrightarrow{P} 0$ and $\eta_n \xrightarrow{d} \xi$, then $\xi_n \xrightarrow{d} \xi$.

Proof: Let F be the cdf of ξ , and let F_n be the cdf of ξ_n . Let t be a continuity point of F , and let $\epsilon > 0$ be such that $t + \epsilon$ and $t - \epsilon$ are also continuity points of F . We write

$$\begin{aligned} F_n(t) &= P\{\xi_n \leq t\} = P\{\xi_n \leq t, |\xi_n - \eta_n| < \epsilon\} + P\{\xi_n \leq t, |\xi_n - \eta_n| \geq \epsilon\} \\ &\leq P\{\eta_n \leq t + \epsilon\} + P\{|\xi_n - \eta_n| \geq \epsilon\}. \end{aligned}$$

By similar reasoning, we obtain

$$P\{\eta_n \leq t - \epsilon\} \leq F_n(t) + P\{|\xi_n - \eta_n| \geq \epsilon\}.$$

As $n \rightarrow \infty$, $P\{\eta_n \leq t \pm \epsilon\} \rightarrow F(t \pm \epsilon)$, and $P\{|\xi_n - \eta_n| \geq \epsilon\} \rightarrow 0$ by the assumption of the theorem. Consequently,

$$F(t - \epsilon) \leq \liminf F_n(t) \leq \limsup F_n(t) \leq F(t + \epsilon).$$

Since $\epsilon > 0$ is arbitrary (subject only to the condition that F is continuous at $\pm\epsilon$), we must have

$$\lim_{n \rightarrow \infty} F_n(t) = F(t),$$

as was to be shown. □

Taking $\eta_n = \xi$, $n = 1, 2, \dots$, we obtain the following theorem:

Theorem 9.5.6 *If $\xi_n \xrightarrow{P} \xi$, then $\xi_n \xrightarrow{d} \xi$.*

Since we already know that convergence a.s. implies convergence in probability, we also have

Theorem 9.5.7 *If $\xi_n \rightarrow \xi$ a.s., then $\xi_n \xrightarrow{d} \xi$.*

Although, as already explained, convergence in distribution does not imply convergence in probability, such implication holds in the case where the convergence is to a constant. We have

Theorem 9.5.8 *If $\xi_n \xrightarrow{d} c$, then $\xi_n \xrightarrow{P} c$.*

Proof: The condition $\xi_n \xrightarrow{d} c$ means that $\{\xi_n\}$ converges in distribution to a random variable ξ such that $P\{\xi = c\} = 1$. Consequently, letting again $F_n(t) = P\{\xi_n \leq t\}$, we have $F_n(x) \rightarrow 0$ for all $x < c$ and $F_n(x) \rightarrow 1$ for all $x > c$. But this means that $P\{|\xi_n - c| > \epsilon\} = P\{\xi_n < c - \epsilon\} + P\{\xi_n > c + \epsilon\} = F_n(c - \epsilon - 0) + [1 - F_n(c + \epsilon)] \rightarrow 0$ for every $\epsilon > 0$, which is what was to be proved. □

We shall state now a theorem that shows the extent to which one is allowed to carry the algebraic manipulations on sequences converging in distribution and in probability.

Theorem 9.5.9 *If ξ_n, α_n , and β_n are sequences of random variables such that $\xi_n \xrightarrow{d} \xi$, $\alpha_n \xrightarrow{P} \alpha$, $\beta_n \xrightarrow{P} \beta$, where α and β are finite constants, then*

$$\alpha_n \xi_n + \beta_n \xrightarrow{d} \alpha \xi + \beta.$$

In particular, we have here the following corollary:

Corollary 9.5.10 If $\xi_n \xrightarrow{d} \xi$, and a, b, a_n, b_n are constants such that $a_n \rightarrow a, b_n \rightarrow b$, then

$$a_n \xi_n + b_n \xrightarrow{d} a\xi + b.$$

The following theorem, which we state without proof, will serve as one of the main tools in proving convergence in distribution to normally distributed random variables:

Theorem 9.5.11 Let $\xi_n, n = 1, 2, \dots$, and ξ be random variables, such that their mgfs $m_n(t)$ and $m(t)$ exist. Then $\xi_n \xrightarrow{d} \xi$ if and only if $m_n(t) \rightarrow m(t)$ for every t in some interval around the point $t = 0$.

The actual use of Theorem 9.5.11 will be based, in most cases, on the assumption of the existence of moments and the corresponding Taylor expansion of the mgf's.

Theorem 9.5.12 Let the random variable X have mgf $m(t)$. Assume that $E(|X|^k) < \infty$ for some k , and let $m_j = E(X^j)$ for $j = 0, 1, \dots, k$. Then

$$m(t) = \sum_{j=0}^k \frac{m_j t^j}{j!} + o(|t|^k), \quad (9.49)$$

where $o(|t|^k)$ is some function such that $o(|t|^k)/|t|^k \rightarrow 0$ as $t \rightarrow 0$.

Proof: The proof consists of using the well-known theorem from calculus on Taylor expansions and recalling (Theorem 7.5.3) that

$$m_j = E(X^j) = \left. \frac{d^j m(t)}{dt^j} \right|_{t=0}.$$

□

The following theorem states another fact known from calculus.

Theorem 9.5.13 If $\{c_n\}$ is a sequence of numbers such that $\lim_{n \rightarrow \infty} c_n = c$, then

$$\lim_{n \rightarrow \infty} \left(1 + \frac{c_n}{n}\right)^n = e^c.$$

As an illustration of applicability of Theorem 9.5.11, we sketch a proof of Theorem 7.8.3 (law of large numbers in the Bernoulli case). We know from (7.22) that the mgf of the binomial random variable S_n is $m_{S_n}(t) = Ee^{tS_n} = (q + pe^t)^n$. Using the fact that $m_{S_n/n}(t) = m_{S_n}(t/n)$, we have

$$m_{S_n/n}(t) = \left[q + p \left(1 + \frac{t}{n} + \frac{t^2}{2n^2} + \dots \right) \right]^n = \left(1 + \frac{pt}{n} + \frac{pt^2}{2n^2} + \dots \right)^n \rightarrow e^{pt}.$$

The last step consists of using Theorem 9.5.13 with

$$c_n = pt + \frac{pt^2}{2n} + \dots \rightarrow pt.$$

We showed that the mgf of S_n/n converges to the mgf of random variable equal identically to p for every t (hence also in some neighborhood of $t = 0$). Theorem 9.5.11 allows us to infer that $S_n/n \xrightarrow{d} p$, and by Theorem 9.5.7, we have $S_n/n \xrightarrow{P} p$.

Weak Laws of Large Numbers

We will now prove some of WLLN using the Chebyshev inequality (Theorem 7.8.1).

Theorem 9.5.14 *Let $X_1, X_2, \dots$ be a sequence of iid random variables. Assume that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2 > 0$. Then, for every $\epsilon > 0$,*

$$\lim_{n \rightarrow \infty} P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| \geq \epsilon \right\} = 0. \quad (9.50)$$

Proof: Based on the Theorem 9.1.1

$$E \left(\frac{X_1 + \dots + X_n}{n} \right) = \mu \quad \text{and} \quad \text{Var} \left(\frac{X_1 + \dots + X_n}{n} \right) = \frac{\sigma^2}{n}.$$

Applying the Chebyshev inequality to variable $(X_1 + \dots + X_n)/n$, we obtain

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| \geq \epsilon \right\} \leq \frac{\sigma^2}{n\epsilon^2}, \quad (9.51)$$

which converges to 0 as $n \rightarrow \infty$. □

This theorem confirms, in some sense, our belief in the “law of averages”: formula (9.51) tells us that if we take the averages $(X_1 + \dots + X_n)/n$ of larger and larger numbers of observations (of some phenomenon, described by random variable X , whose “copies” $X_1, X_2, \dots$ are being observed), then it becomes less and less likely that the average $(X_1 + \dots + X_n)/n$ deviates by more than ϵ from the “true average” $\mu = E(X)$.

This assertion of Theorem 9.5.14 constitutes, to a large extent, the basis of the common understanding why “statistics works,” that is, why increasing the number of observations pays off in the form of being able to make a better inference about some quantities.

The generality of the assertion of Theorem 9.5.14 is in stark contrast with narrowness of the assumptions specifying that $X_1, X_2, \dots$ are iid random variables. The importance—practical and cognitive—of assertion (9.50) makes it worthwhile to analyze to which extent that assumption of Theorem 9.5.14 can be relaxed.

The full answer lies beyond the scope of this book. We will, however, analyze the question of to what extent the assumptions can be relaxed under the present proof, based on the Chebyshev inequality.

First, let us observe that we did not fully utilize the assumption that X_i ’s have the same distribution. In fact we used only a special consequence of this assumption, namely the fact that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$ are the same for all i . Thus, we can drop the requirement that X_i ’s be identically distributed, as long as we retain the stationarity of the mean and variance. To use the example of measurement, we could take measurements of the same quantity μ with different measuring instruments or methods, provided that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$ (such measurements are called unbiased and having the same precision).

But even the requirement that $E(X_i) = \mu$ is not necessary: when $E(X_i) = \mu_i$, relation (9.50) can be replaced by

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \frac{\mu_1 + \dots + \mu_n}{n} \right| \geq \epsilon \right\} \rightarrow 0, \quad (9.52)$$

which is the same as

$$P \left\{ \left| \frac{U_1 + \dots + U_n}{n} \right| \geq \epsilon \right\} \rightarrow 0,$$

where $U_i = X_i - \mu_i$ is the deviation of the i th random variable from its own mean.

Next, an inspection of the proof of Theorem 9.5.14 shows that it is not necessary that variances all be equal. What is required is that the variance of the average $(X_1 + \cdots + X_n)/n$ decreases to 0 as n increases. We can therefore formulate the following version of the law of large numbers:

Theorem 9.5.15 *Let $X_1, X_2, \dots$ be iid random variables, with $E(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$. If*

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2 = 0, \quad (9.53)$$

then relation (9.52) holds for every $\epsilon > 0$.

Let us now see to what extent it is possible to relax the assumption of independence. Again, an inspection of the proof of Theorem 9.5.14 shows that the property which was used is the additivity of variance, specifically the fact that $\text{Var}(X_1 + \cdots + X_n) = \sigma_1^2 + \cdots + \sigma_n^2$. But this property holds under the assumption that the random variables are uncorrelated.

Theorem 9.5.16 *If random variables $X_1, X_2, \dots$ are uncorrelated then the condition (9.53) is sufficient for $(1/n) \sum_{i=1}^n (X_i - \mu_i) \xrightarrow{P} 0$.*

Finally, let us observe that even the latter condition can be relaxed: what we really need to make the proof valid is that $(1/n^2) \text{Var}(X_1 + \cdots + X_n) \rightarrow 0$. This in turn is implied by the assumption (9.53) in the case where all covariances are zero or negative. We therefore have

Theorem 9.5.17 *Let $X_1, X_2, \dots$ be random variables with $E(X_i) = \mu_i$ and $\sigma_j^2 = \text{Var}(X_j)$, and such that $\text{Cov}(X_i, X_j) \leq 0$ for $i \neq j$, satisfying (9.53) as $n \rightarrow \infty$. Then for every $\epsilon > 0$,*

$$P \left\{ \left| \frac{1}{n} \sum_{j=1}^n (X_j - \mu_j) \right| \geq \epsilon \right\} \rightarrow 0$$

as $n \rightarrow \infty$.

This is as far as we will go using the techniques of proof based on the Chebyshev inequality. It is of some interest that in the case of identical distributions, one can prove a weak law of large numbers without the assumption of existence of variance.

Theorem 9.5.18 *Let $X_1, X_2, \dots$ be a sequence of iid random variables with $E(X_i) = \mu$, and such that their common mgf exists in some neighborhood of $t = 0$. Then the relation (9.50) holds.*

Proof: From Theorem 9.5.7, it is enough to show that $(X_1 + \cdots + X_n)/n \xrightarrow{d} \mu$, and therefore (Theorem 9.5.11) that $m_{S_n/n}(t) \rightarrow e^{\mu t}$ in some neighborhood of $t = 0$.

Letting $m(t) = Ee^{tX}$, we have $m_{S_n/n} = [m(t/n)]^n$, since X_i 's are iid. The existence of $\mu = E(X)$ implies the existence of the derivative m' , and the relation $m'(0) = \mu$. Then, the Taylor expansion of $m(t)$ is $m(t) = 1 + \mu t + o(|t|)$ (see Theorem 9.5.12) so that

$$m_{S_n/n}(t) = \left(1 + \frac{\mu t}{n} + o\left(\frac{|t|}{n}\right) \right)^n.$$

To complete the proof, we use Theorem 9.5.13 for $c_n = \mu t + no(|t|/n) \rightarrow \mu t$. □

Comment. Observe that the assumption of existence of an mgf is not necessary for the validity of the theorem. The proof of this strengthened version requires nothing more than replacing mgfs with characteristic functions. All the steps of the proof remain unchanged.

Strong Laws of Large Numbers

The strong laws of large numbers (SLLNs) are theorems that assert almost sure convergence of sequences of random variables obtained by averaging some underlying sequences of random variables. We will prove here two such theorems, both due to Kolmogorov.

Theorem 9.5.19 *Let $X_1, X_2, \dots$ be independent, with $EX_i = \mu_i$, $X_i = \sigma_i^2$. If*

$$\sum_{n=1}^{\infty} \frac{\sigma_n^2}{n^2} < \infty,$$

then

$$\frac{1}{n} \sum_{i=1}^n (X_i - \mu_i) \rightarrow 0 \quad \text{a.s.} \quad (9.54)$$

Proof: To simplify notation, let $S_n = \sum_{j=1}^n X_j$ and $m_n = \sum_{i=1}^n \mu_i$. For a fixed $\epsilon > 0$, let

$$C_k = \left\{ \max_{2^{k-1} < n \leq 2^k} \frac{1}{n} |S_n - m_n| \geq \epsilon \right\}.$$

To prove the theorem, it suffices to show that $\sum P(C_k) < \infty$, since then, by the Borel–Cantelli lemma (Theorem 2.6.2), only finitely many events C_k will occur a.s., which implies (9.54).

If C_k occurs, then at least one of the inequalities

$$|S_n - E(S_n)| \geq \epsilon n \geq \epsilon 2^{k-1}, \quad n = 2^{k-1} + 1, \dots, 2^k,$$

occurs. By the Kolmogorov inequality (Theorem 7.8.5), we obtain

$$P(C_k) \leq \frac{\text{Var}(S_{2^k})}{\epsilon^2 (2^{2k-2})} = \frac{4}{\epsilon^2} \times \frac{\sum_{j=1}^{2^k} \sigma_j^2}{2^{2k}}.$$

We can therefore write

$$\begin{aligned} \sum_{k=1}^{\infty} P(C_k) &\leq \frac{4}{\epsilon^2} \sum_{k=1}^{\infty} \frac{1}{2^{2k}} \sum_{j=1}^{2^k} \sigma_j^2 = \frac{4}{\epsilon^2} \sum_{j=1}^{\infty} \sigma_j^2 \sum_{2^k \geq j} \frac{1}{2^{2k}} \\ &= \frac{16}{3\epsilon^2} \sum_{j=1}^{\infty} \frac{\sigma_j^2}{j^2} < \infty, \end{aligned}$$

since

$$\sum_{2^k \geq j} \frac{1}{2^{2k}} = \sum_{k \geq \log_2 j} 2^{-2k} = \frac{2^{-2\log_2 j}}{1 - \frac{1}{4}}.$$

□

We will now prove another strong law of large numbers, also due to Kolmogorov, that covers the iid case.

Theorem 9.5.20 *Let $X_1, X_2, \dots$ be iid random variables, and let $S_n = X_1 + \dots + X_n$. If $\mu = E(X_i)$ exists, then $S_n/n \rightarrow \mu$ a.s.*

Proof. Let us truncate the random variables $\{X_n\}$, by letting

$$Y_k = \begin{cases} X_k, & \text{if } |X_k| \leq k, \\ 0, & \text{if } |X_k| > k. \end{cases}$$

We have

$$\sum_{k=1}^{\infty} P\{Y_k \neq X_k\} = \sum_{k=1}^{\infty} P\{|X_k| > k\} = \sum_{k=1}^{\infty} P\{|X_1| > k\} < \infty$$

by the assumption that $E(X_1) < \infty$ and since X_k has the same distribution as X_1 . Thus, the inequality $X_k \neq Y_k$ will occur a.s. only a finite number of times, and it suffices to prove that $(1/n) \sum_{k=1}^n Y_k \rightarrow \mu$ a.s. To this end, we will use Theorem 9.5.19, proved above. We let $\sigma_k^2 = \text{Var}(Y_k) \leq E(Y_k^2)$. Therefore, we can write

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{\sigma_k^2}{k^2} &\leq \sum_{k=1}^{\infty} \frac{E(Y_k^2)}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} \int_{-k}^k x^2 dF(x) \\ &= \sum_{k=1}^{\infty} \frac{1}{k^2} \sum_{j=1}^k \int_{j-1 < |x| \leq j} x^2 dF(x) \\ &= \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} x^2 dF(x) \sum_{k=j}^{\infty} \frac{1}{k^2} \leq \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} x^2 dF(x) \times \frac{C}{j} \\ &\leq C \sum_{j=1}^{\infty} \int_{j-1 < |x| \leq j} |x| dF(x) = C \int_{-\infty}^{\infty} |x| dF(x) < \infty. \end{aligned}$$

We use the estimate $1/j^2 + 1/(j+1)^2 + \cdots < C/j$ valid for some C . It follows that $(1/n) \sum_{j=1}^n [Y_j - E(Y_j)] \rightarrow 0$ a.s. But $E(Y_j) \rightarrow E(X_1) = \mu$ as $j \rightarrow \infty$; hence also $(1/n)[E(Y_1) + \cdots + E(Y_n)] \rightarrow \mu$. Then, we must have

$$\frac{1}{n} \sum_{j=1}^n Y_j \rightarrow \mu \quad \text{a.s.},$$

which completes the proof. □

PROBLEMS

9.5.1 Let $X_1, \dots, X_n$ be a random sample from a $\text{POI}(\lambda)$ distribution. Show that

$$e^{-\bar{X}_n} \xrightarrow{P} P(X_1 = 0).$$

9.5.2 Let $Z_1, \dots, Z_n$ be a random sample from $N(0, 1)$ distribution. Find the limiting distribution of $Y_n = \sum_{i=1}^n (Z_i + 1/n)/\sqrt{n}$.

9.5.3 Let $X_1, \dots, X_n$ be a random sample from distribution with cdf $F(x) = 1 - x^{-2}$ for $1 \leq x < \infty$, and 0 otherwise. Find the cdf of limiting distribution of: **(i)** $X_{1:n}$. **(ii)** $X_{1:n}^n$. **(iii)** $X_{n:n}/\sqrt{n}$.

9.5.4 Let $X_1, \dots, X_n$ be a random sample from continuous distribution with a cdf F such that F^{-1} exists. Find (if it exists) the limiting distribution of: **(i)** $U_n = nF(X_{1:n})$. **(ii)** $W_n = n[1 - F(X_{n:n})]$. **(iii)** $V_n = nF(X_{3:n})$.

9.5.5 Let $X_1, \dots, X_n$ be a random sample from distribution with cdf $F(x) = x^2/9$ for $0 < x < 3$. Find the limiting distribution of $Y_n = \sqrt{n}X_{1:n}$ if it exists.

9.5.6 A random variable has a Pareto distribution with parameters a, b ($a > 0, b > 0$) if its density is

$$f(x; a, b) = \frac{a}{b(1 + x/b)^{a+1}}, \quad x > 0.$$

Let $X_1, \dots, X_n$ be a random sample from the Pareto distribution with density $f(x; 1, 1)$. **(i)** Find the limiting distribution of random variable $U_n = nX_{1:n}$. [*Hint:* Find cdf $F(x)$ of X_i , and then determine cdf of U_n in terms of $F(x)$.] **(ii)** Show that $V_n = X_{n:n}$ does not have a proper limiting distribution; specifically, $\lim_{n \rightarrow \infty} P\{V_n \leq t\} = 0$ for every t . **(iii)** Find the limiting distribution of V_n/n .

9.5.7 Let $X_1, \dots, X_n$ be a random sample from a logistic distribution with a cdf $F(x) = 1/(1 + e^{-x})$, and let $V_n = X_{n:n}$. Then $V_n \xrightarrow{P} \infty$, but $V_n - \log n$ has a proper limiting distribution. Find $\lim_{n \rightarrow \infty} P\{V_n - \log n \leq 0\}$ and $\lim_{n \rightarrow \infty} P\{|V_n - \log n| \leq 1\}$.

9.6 CENTRAL LIMIT THEOREM

The term *central limit theorem* (CLT) is a generic name used to designate any theorem that asserts that the sums of large numbers of random variables, after standardization (i.e., subtraction of the mean and division by standard deviation), have approximately a standard normal distribution.

As suggested by the adjective *central*, the search for conditions under which sums of a large number of components have an approximate normal distribution has been (and to a large extent still is) one of the leading research topics in probability theory for the last 200 years or so. We begin with the simplest case of iid sequences.

Theorem 9.6.1 (Lindeberg and Lévy) Let $X_1, X_2, \dots$ be a sequence of iid random variables with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, where $0 < \sigma^2 < \infty$. Then, letting $S_n = X_1 + \dots + X_n$, for every x

$$\lim_{n \rightarrow \infty} P\left\{\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt. \quad (9.55)$$

Proof: Observe that $E(S_n) = n\mu$ and $\text{Var}(S_n) = n\sigma^2$, so the left-hand side of (9.55) is simply the limit of the cdf's of the standardized sums

$$S_n^* = \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}}.$$

The right-hand side is the cdf of a standard normal random variable, denoted $\Phi(x)$. Thus, the theorem asserts that the cdf's of S_n^* converge to $\Phi(x)$ for every x , hence at every point of continuity of $\Phi(x)$. Letting Z denote the standard normal random variable, Theorem 9.6.1 asserts that $S_n^* \xrightarrow{d} Z$.

When the assertion is phrased in this way, it should be clear that a possible strategy of the proof is to use a mgf and Theorem 9.5.11. Let $m_X(t)$ be the common mgf of random variables X_i (assumed to exist). The existence of the first two moments suggests applying a Taylor expansion. We have

$$S_n^* = \sum_{i=1}^n \frac{X_i - \mu}{\sigma\sqrt{n}}$$

so that

$$m_{S_n^*}(t) = [m_{(X-\mu)/\sigma\sqrt{n}}(t)]^n = \left[m_{(X-\mu)/\sigma} \left(\frac{t}{\sqrt{n}} \right) \right]^n. \quad (9.56)$$

Now

$$E \left(\frac{X-\mu}{\sigma} \right) = 0, \quad E \left(\frac{X-\mu}{\sigma} \right)^2 = 1 \quad (9.57)$$

so that

$$m_{(X-\mu)/\sigma}(t) = 1 + \frac{t^2}{2} + o(t^2).$$

Consequently, using (9.56), we obtain

$$m_{S_n^*} = \left[1 + \frac{t^2}{2n} + o \left(\frac{t^2}{n} \right) \right]^n = \left[1 + \frac{t^2/2 + no(t^2/n)}{n} \right]^n.$$

By Theorem 9.5.13, for $c_n = \frac{t^2}{2} + no(\frac{t^2}{n}) \rightarrow \frac{t^2}{2}$ we obtain

$$m_{S_n^*}(t) \rightarrow e^{t^2/2},$$

which completes the proof in the special case where the underlying random variables X_i have mgf's. For the general case, see the comment following the proof of Theorem 9.5.18. $\square$

■ EXAMPLE 9.12

Suppose that you buy a supply of 20 household batteries to be used for some specific purpose, one after another when needed. Assume that unused batteries are not aging and that the lifetime of each such battery is a random variable with a mean of two weeks and a standard deviation of three days. What is the (approximate) probability that the supply of batteries will last more than 9 but less than 10 months? (i.e., more than 270 and fewer than 300 days?).

SOLUTION. The question here is about the probability that $S_{20} = X_1 + \cdots + X_{20}$ satisfies the inequality $270 < S_{20} < 300$. We have $n\mu = 20 \times 14 = 280$ and $\sigma\sqrt{n} = 3 \times \sqrt{20} = 13.4$. Thus, $270 < S_{20} < 300$ occurs if and only if $(270 - 280)/13.4 < S_{20}^* < (300 - 280)/13.4$, (i.e., if $-0.75 < S_{20}^* < 1.49$).

For $Z \sim N(0, 1)$, we have

$$\begin{aligned} P\{270 < S_{20} < 300\} &\approx P\{-0.75 < Z < 1.49\} = \Phi(1.49) - \Phi(-0.75) \\ &= 0.9319 - 0.2266 = 0.7053. \end{aligned}$$

Theorem 9.6.1 is a generalization of the oldest central limit theorem, due to Laplace, covering the binomial distribution.

Theorem 9.6.2 Laplace *If S_n has a binomial distribution $\text{BIN}(n, p)$, then for any $z_1 < z_2$,*

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{A_n \leq j \leq B_n} P\{S_n = j\} &= \lim_{n \rightarrow \infty} P\{z_1 \leq S_n^* \leq z_2\} \\ &= \Phi(z_2) - \Phi(z_1), \end{aligned} \quad (9.58)$$

where $A_n = np + z_1\sqrt{npq}$, $B_n = np + z_2\sqrt{npq}$.

Proof: If S_n is the number of successes in n independent trials, each with probability of success p , then $S_n = X_1 + \cdots + X_n$, where $P\{X_i = 1\} = 1 - P\{X_i = 0\} = p$, $i = 1, 2, \dots, n$. Consequently, $E(S_n) = np$ and $\text{Var}(S_n) = npq$ so that

$$S_n^* = \frac{S_n - np}{\sqrt{npq}}.$$

□

Suppose now that we want to find $P\{a \leq S_n \leq b\}$. According to Theorem 9.6.2, for large n we have

$$\begin{aligned} P\{a \leq S_n \leq b\} &= P\left\{\frac{a - np}{\sqrt{npq}} \leq S_n^* \leq \frac{b - np}{\sqrt{npq}}\right\} \\ &\approx \Phi\left(\frac{b - np}{\sqrt{npq}}\right) - \Phi\left(\frac{a - np}{\sqrt{npq}}\right). \end{aligned} \quad (9.59)$$

However, approximation (9.59) can be improved somewhat if we observe that in the present case S_n is an integer-valued random variable, and for integers a and b the exact expression is

$$P\{a \leq S_n \leq b\} = \sum_{j=a}^b P\{S_n = j\}. \quad (9.60)$$

Each term on the right-hand side of (9.60) can be approximated by an area under the normal curve between $(j - 0.5 - np)/\sqrt{npq}$ and $(j + 0.5 - np)/\sqrt{npq}$.

Adding such approximations, the terms for neighboring j cancel, and we obtain the following formula:

$$P\{a \leq S_n \leq b\} \approx \Phi\left(\frac{b + 0.5 - np}{\sqrt{npq}}\right) - \Phi\left(\frac{a - 0.5 - np}{\sqrt{npq}}\right) \quad (9.61)$$

for any integers $a \leq b$, known as a *normal approximation with continuity correction*.

■ EXAMPLE 9.13

A fair coin is tossed $n = 15$ times. Find the approximate probability that the number S_{15} of heads will satisfy the inequality $8 \leq S_{15} < 10$.

SOLUTION. Notice that $8 \leq S_{15} < 10$ is the same inequality as $8 \leq S_{15} \leq 9$. Since $np = 15 \times 0.5 = 7.5$, $\sqrt{npq} = \sqrt{15 \times 0.5 \times 0.5} = 1.94$, the approximation (9.58) gives

$$\begin{aligned} P\{8 \leq S_{15} \leq 9\} &\approx \Phi\left(\frac{9 + 0.5 - 7.5}{1.94}\right) - \Phi\left(\frac{8 - 0.5 - 7.5}{1.94}\right) \\ &= \Phi(1.03) - \Phi(0) = 0.8485 - 0.5 = 0.3485. \end{aligned}$$

The exact value is

$$\begin{aligned} P\{S_{15} = 8\} + P\{S_{15} = 9\} &= \binom{15}{8} \left(\frac{1}{2}\right)^{15} + \binom{15}{9} \left(\frac{1}{2}\right)^{15} \\ &= 0.1964 + 0.1527 = 0.3491. \end{aligned}$$

Generally, the quality of approximation improves as n increases, and—for the same n —decreases as p moves away from $1/2$ in either direction.

■ EXAMPLE 9.14 Decision Problem

Suppose that we design a theater with 1,000 seats. The theater has two entrances, A and B, situated with respect to a parking lot, public transportation, and so on, so that the patrons have equal chances of choosing any of the entrances.

Suppose also that our theater is to be located in a city where the climate calls for patrons to wear overcoats, which they can leave in a coatroom. There are to be two coatrooms, each located near one of the entrances, and while it is possible to enter through one entrance and leave the coat in a coatroom near to the other entrance, it is inconvenient to do so. How many coat hangers should each coatroom have?

SOLUTION. Clearly, the problem is not precise enough as stated: We have to specify the criterion which we want to attain. One of the objectives is minimization of the cost of equipping the coatroom in hangers, racks, and so on. We do not want to staff a coatroom that will remain empty. On the other hand, we do not want to inconvenience patrons by making them go to a distant coatroom. The two extremes, each satisfying one of the foregoing objectives, is to equip each coatroom with 1,000 coat hangers, and to equip each with exactly 500 coat hangers.

A possible objective may be: We want to equip each coatroom with $500 + x$ hangers, where x is the smallest number such that (say) on 95% of nights when the theater is sold out, everyone will be able to leave his or her coat at the coatroom nearest to the entrance used.

To solve the problem, we have to make some assumptions about independence of choice of entrances A and B by the patrons. As the first approximation, assume that the patrons arrive one at a time and each chooses the entrance independent of other patrons. Let $S_{1,000}$ be the number of patrons (among $n = 1,000$) who choose entrance A. We want to have $S_{1,000} \leq 500 + x$ (then everyone who enters through A is not inconvenienced), and also $1,000 - S_{1,000} \leq 500 + x$ (which is the analogous condition for entrance B). Thus, we would like the event $500 - x \leq S_{1,000} \leq 500 + x$ to occur with probability 0.95 or more. Since $p = 0.5$, $np = 1,000 \times 0.5 = 500$, and $\sqrt{npq} = \sqrt{1,000 \times 0.5 \times 0.5} = 15.8$, we have

$$\begin{aligned} P\{500 - x \leq S_{1,000} \leq 500 + x\} &= P\left\{\frac{-x - 0.5}{15.8} \leq S_{1,000}^* \leq \frac{x + 0.5}{15.8}\right\} \\ &\approx \Phi\left(\frac{x + 0.5}{15.8}\right) - \Phi\left(-\frac{x + 0.5}{15.8}\right) \geq 0.95. \end{aligned}$$

Since $\Phi(1.96) - \Phi(-1.96) = 0.95$, we must therefore take x as the smallest integer for which $(x + 0.5)/15.8 \geq 1.96$, which gives $x = 31$. Thus, to achieve our objective, we should install 532 coat hangers in each coatroom.

■ EXAMPLE 9.15

Continuing Example 9.14, a more realistic assumption is that people attend the theater in pairs, and both members of a pair come through the same entrance. We have now $n = 500$ pairs, and letting S_{500} denote the number of pairs who choose entrance A, we must have $2S_{500} \leq 500 + x$ and $1,000 - 2S_{500} \leq 500 + x$. Now $E(S_{500}) = 500 \times 0.5 = 250$ and $\text{Var}(S_{500}) = 500 \times 0.5 \times 0.5 = 125$. Therefore

$250 - x/2 \leq S_{500} \leq 250 + x/2$. Using formula (9.58), we get

$$\begin{aligned} P\left\{250 - \frac{x}{2} \leq S_{500} \leq 250 + \frac{x}{2}\right\} &= P\left\{\frac{-\frac{x}{2} - 0.5}{\sqrt{125}} \leq S_{500}^* \leq \frac{\frac{x}{2} + 0.5}{\sqrt{125}}\right\} \\ &= \Phi\left(\frac{x+1}{22.3}\right) - \Phi\left(-\frac{x+1}{22.3}\right). \end{aligned}$$

Again, x is the smallest integer for which $(x+1)/22.3 \geq 1.96$, so $x = 43$. We see that grouping (in this case into pairs, but the effect is the same for other groupings) a set of persons, with groups choosing the entrance independently, increases the variability: we now need to supply 543 hangers in each coatroom to meet the requirement.

The CLT proved thus far concerns the rather narrow case of independent and identically distributed components. In this case, the sum has asymptotically normal distribution, provided only that variance is finite. This theorem is often utilized to explain the frequent appearance of a normal distribution in nature. The argument typically goes along the lines of attribute, such as the height of a person, or an error of measurement of some quantity, (such as the speed of light). Whatever the case, the observed value is influenced by a large number of factors, some having a negative and some having a positive effect. Some such factors are known, but their effect cannot be predicted exactly, whereas other factors may not even be named. What matters is that all these factors operate largely independently of one another and each in isolation is small as compared with the total effect of all factors (i.e., factors that are known to have large effects are treated differently and are not included in these considerations). The CLT therefore asserts that the total effect of such “small” factors is random and has approximately a normal distribution. We already know this to be true in case of factors that are iid. However, independence and identical distribution can hardly be justified in every real situation, and can at best be regarded as approximations to reality. Consequently, one can expect central limit theorems to be valid in wider classes of situations where the iid assumption does not hold. We state below a number of theorems that provide conditions for asymptotic normality in case of independent random variables that are not identically distributed.

Theorem 9.6.3 (Liapunov) *Let $X_1, X_2, \dots$ be a sequence of random variables such that $E(X_i) = \mu_i$, $\text{Var}(X_i) = \sigma_i^2$, and $\gamma_i = E|X_i - \mu_i|^3$. Moreover, put*

$$m_n = \sum_{j=1}^n \mu_j, \quad s_n^2 = \sum_{j=1}^n \sigma_j^2, \quad \Gamma_n = \sum_{j=1}^n \gamma_j,$$

and let $S_n = X_1 + X_2 + \dots + X_n$ be the corresponding sequence of partial sums. If additionally $X_1, X_2, \dots$ have finite third moments, then the condition

$$\lim_{n \rightarrow \infty} \frac{\Gamma_n}{s_n^3} = 0 \tag{9.62}$$

is sufficient for convergence

$$\frac{S_n - m_n}{s_n} \xrightarrow{d} N(0, 1).$$

The proof of this theorem lies beyond the scope of this book.

As an illustration of the application of the Liapunov theorem, consider a sequence of independent trials, such that in n th trial the probability of success is p_n . We let S_n denote the number of successes in the n first trials so that $S_n = X_1 + \dots + X_n$, where $X_i = 1$ or 0 depending

on whether or not the i th trial leads to a success. We then have $E(X_i) = p_i$, $\text{Var}(X_i) = p_i q_i$, while the third absolute central moment γ_i is

$$\gamma_i = E|X_i - p_i|^3 = |1 - p_i|^3 P(X_i = 1) + |0 - p_i|^3 P(X_i = 0) = q_i^3 p_i + p_i^3 q_i.$$

Thus $\gamma_i = p_i q_i (p_i^2 + q_i^2) \leq p_i q_i$, and

$$\frac{\Gamma_n}{s_n^3} \leq \frac{\sum_{i=1}^n p_i q_i}{(\sum_{i=1}^n p_i q_i)^{3/2}} = \frac{1}{(\sum_{i=1}^n p_i q_i)^{1/2}}.$$

Consequently, the Liapunov condition (9.62) holds if $\sum_{i=1}^n p_i q_i = \infty$, and we have

Theorem 9.6.4 *Consider a sequence of independent trials, with the probability of success in the i th trial being p_i . If S_n is the number of successes in the n first trials, then*

$$\frac{S_n - \sum_{i=1}^n p_i}{\sqrt{\sum_{i=1}^n p_i q_i}} \xrightarrow{d} N(0, 1)$$

if $\sum_{i=1}^{\infty} p_i q_i = \infty$.

We close this chapter by stating a theorem that completed the long search for conditions implying limiting normality in case of independent random variables (the cases of dependent random variables are still the object of intense research).

Theorem 9.6.5 (Lindeberg and Feller) *Let $X_1, X_2, \dots$ be a sequence of independent random variables with finite second moments. Assume that $s_n^2 \rightarrow \infty$ and $\max_{1 \leq j \leq n} \sigma_j^2 / s_n^2 \rightarrow 0$ as $n \rightarrow \infty$. Then,*

$$\frac{S_n - m_n}{s_n} \xrightarrow{d} N(0, 1)$$

if and only if for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^2} \sum_{j=1}^n \int_{|x - \mu_j| \geq \epsilon s_n} (x - \mu_j)^2 dF_j(x) = 0, \quad (9.63)$$

where F_j is the cdf of X_j .

The “if” part was proved by Lindeberg, so (9.63) is called the Lindeberg condition. The “only if” part is due to Feller.

Let us now introduce the notation useful in describing rates of convergence in probability. Let $V_1, V_2, \dots$ be a sequence of random variables. The symbol $o_P(V_n)$ denotes random variables of order smaller than V_n for large n , in other words such that $o_P(V_n)/V_n \xrightarrow{P} 0$. Similarly, the symbol $O_P(V_n)$ represents a sequence of variables of the same order of convergence as V_n , so that the sequence $O_P(V_n)/V_n$ converges in probability to a constant ($O_P(V_n)/V_n \xrightarrow{P} C$).

The following theorem is a generalization of a CLT that is very useful when only the first two moments, and not the entire probability distribution, are known. The theorem provides an approximate distribution of the transformed random variables obtained in random samples.

Theorem 9.6.6 (Delta Method) *Let $X_1, X_2, \dots$ be a sequence of random variables such that*

$$\sqrt{n}(X_n - \theta) \rightarrow N(0, \sigma^2), \quad (9.64)$$

and let g be a function with a nonzero derivative $g'(\theta)$. Then

$$\sqrt{n}[g(X_n) - g(\theta)] \rightarrow N(0, \sigma^2[g'(\theta)]^2). \quad (9.65)$$

Proof: The Taylor expansion of $g(X_n)$ around $X_n = \theta$ gives

$$g(X_n) = g(\theta) + g'(\theta)(X_n - \theta) + o_P(|X_n - \theta|). \quad (9.66)$$

Rearranging (9.66) and multiplying by $\sqrt{n}$, we obtain

$$\begin{aligned} \sqrt{n}(g(X_n) - g(\theta)) &= g'(\theta)\sqrt{n}(X_n - \theta) + \sqrt{n} o_P(|X_n - \theta|), \\ \sqrt{n}(g(X_n) - g(\theta)) &= g'(\theta)\sqrt{n}(X_n - \theta) + o_P(\sqrt{n}|X_n - \theta|). \end{aligned} \quad (9.67)$$

Based on (9.64), convergence of X_n to θ in probability, and Theorem 9.5.5, we obtain (9.65). $\square$

Let us now illustrate the Delta method by an example.

■ EXAMPLE 9.16

Let X_n be a relative frequency of a success in n Bernoulli trials. Find the asymptotic distribution of $g(X_n) = 1/X_n$.

SOLUTION. Let p be the probability of success in a single trial. Then $g'(p) = -1/p^2$. Consequently, based on Theorems 9.6.5, 9.5.6, and the fact that

$$\sqrt{n}(X_n - p) \rightarrow N(0, p(1 - p)),$$

we immediately obtain

$$\sqrt{n}\left(\frac{1}{X_n} - \frac{1}{p}\right) \rightarrow N\left(0, \frac{(1 - p)}{p^3}\right).$$

Last we would like to mention two possible extensions of the Delta method. If $g'(\theta) = 0$, then one can take one more term in the Taylor expansion (9.66). Such approach will yield a second-order Delta method (for details see e.g., Casella and Berger, 2002). Also, the Delta method can be easily extended to a multivariate setting (see Oehlert, 1992).

PROBLEMS

9.6.1 Let $X_1, \dots, X_n$ be a random sample from the distribution with density $f(x) = xe^{-x}$, $x > 0$. Find c if it is known that $P\{\bar{X}_n > c\} = 0.75$ for $n = 250$.

9.6.2 Assume that 500 students at a certain college will graduate on a given day. Because of space limitations, the college offers each student two tickets for the commencement ceremony. From past experience, it is known that 50% of the students will invite two guests to attend the ceremony, 20% students will invite one guest, and the remaining 30% will not attend at all, so they will invite no guests. How many chairs should the college order to have at least 95% chance that all attending guests will have seats?

- 9.6.3** Passengers on an international flight have a luggage weight limit B . The actual weight W of the passenger's luggage is such that W/B has a $\text{BETA}(a, b)$ distribution where $a/(a+b) = 0.9$. Assume that the weights of luggage of different passengers are independent and that the plane has 220 seats. Find a and b if it is known that when the plane is fully booked, then there is a 5% chance that the total weight of baggage will exceed $200B$.
- 9.6.4** Let $X_1, \dots, X_n$ be a random sample from the $\text{BETA}(4, 2)$ distribution. Let $S_n = X_1 + \dots + X_n$. Find the smallest n for which $P\{S_n \geq 0.75n\} \leq 0.01$.
- 9.6.5** A regular dodecahedron (12-sided Platonian solid) has six red and six white faces, with the faces of each color labeled 1, ..., 6. If you toss a face with label k you pay or win $\$k$, depending on whether the color is red or white. Find the probability that after 50 tosses you are ahead by more than $\$10$.
- 9.6.6** Let $X_1, \dots, X_{360}$ represent the outcomes of 360 tosses of a fair die. Let S_{360} be the total score $X_1 + \dots + X_{360}$, and for $j = 1, \dots, 6$, let Y_j be the total number of tosses that give outcome j . Use normal approximation to obtain: (i) $P(55 < Y_3 < 62)$. (ii) $P(1,200 < S_{360} < 1,300)$. (iii) $P(1,200 < S_{360} < 1,300 | Y_1 = 55)$. (iv) $P(1,200 < S_{360} < 1,300 | Y_4 = 55)$. (v) $P(1,200 < S_{360} < 1,300 | X_1 = X_2 = \dots = X_{55} = 4)$. (vi) Check if continuity correction improves approximations in (i)–(v).
- 9.6.7** A fair coin is tossed $2n$ times. How large must n be if it is known that the probability of the equal number of heads and tails is less than 0.1?
- 9.6.8** A rate at which patients arrive to an urgent care center in the evening is one per 6 minutes. Assuming that the number of arrivals follows Poisson distribution, find the probability that more than 35 patients but no more than 50 will arrive to the center on a given day between 6 p.m. and 10 p.m. Use normal approximation with continuity correction.
- 9.6.9** Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(\lambda)$ distribution. Show that there exist normalizing constants A_n and B_n such that $(X_1 + \dots + X_n - A_n)/B_n$ has asymptotically $N(0, 1)$ distribution.
- 9.6.10** Referring to Example 7.39, assume that a man's shoe has an average length of 1 foot and $\sigma = 0.1$ foot. Find the (approximate) probability that the mean of 16 lengths of men's shoes exceed 1 foot by more than 1 inch.
- 9.6.11** A die is unbalanced in such a way that the probability of tossing k ($k = 1, \dots, 6$) is proportional to k . You pay $\$4$ for a toss, and win $\$k$ if you toss k . Find the approximate probability that you are ahead after 100 tosses.
- 9.6.12** Let $\bar{X}_n$ be a sample mean in a random sample $X_1, \dots, X_n$ from $\text{POI}(\lambda)$ distribution and let $T_n = \sum_{i=1}^n X_i$. Use the Delta method to find the limiting distribution of: (i) $g(\bar{X}_n) = \sqrt{n}(\bar{X}_n - \lambda)$. (ii) $g(\bar{X}_n) = 1/\bar{X}_n$. (iii) $g(T_n) = \log(T_n)$. (iv) $g(T_n) = \sqrt{T_n}$. [Hint: T_n/n has approximately $N(\lambda, \lambda/n)$ distribution (why?).]
- 9.6.13** Let $\bar{X}_n$ be a relative frequency of success in n Bernoulli trials. Use the Delta method to find the limiting distribution of: (i) $g(\bar{X}_n) = \bar{X}_n(1 - \bar{X}_n)$. (ii) $g(\bar{X}_n) = \sqrt{\bar{X}_n(1 - \bar{X}_n)}$. (iii) $g(\bar{X}_n) = \sqrt{(1 - \bar{X}_n)/\bar{X}_n}$.